The sharp convergence range and boundary rates of generalized Fast Krasnoselskii–Mann iteration

Abstract

We study the generalized Fast Krasnoselskii–Mann iteration for a nonexpansive mapping on a real Hilbert space. The known fast parameter interval is $1 < \theta < \alpha - 1$, with $\alpha > 2$. We determine its universal convergence range for positive θ and distinguish this range from the acceleration boundary.

- For each $\alpha > 1$, weak convergence for every nonexpansive map with a fixed point, every $\sigma > 0$ and every initial pair holds exactly when $1 \le \theta < \alpha$. The case $\theta = 1$ is the known Halpern boundary.
- At $\theta = \alpha - 1$ with $\alpha > 2$, the residual and velocity are $O(1/k)$. In the higher band $\max\{1, \alpha - 1\} < \theta < \alpha$, their sharp general order is $O(k^{-(\alpha-\theta)})$.
- Below $\theta = 1$, one fixed nonlinear firmly nonexpansive map has an unbounded orbit for each parameter pair, with $\sigma = \alpha$. In finite dimensions, strictly averaged maps with nonempty bounded fixed-point sets instead approach that set. Reflections give the upper failure and exact rate sharpness.
- At the corresponding continuous acceleration boundary, monotone Lipschitz operators give weak trajectory convergence and $O(1/t)$ residual and velocity bounds.

For a convex proximal map with a fixed point, residuals and steps vanish for all positive α, θ and σ; a finite-dimensional unique minimizer gives strong convergence. We use two-state energies and scalar averaging, and prove common finite limits for scaled residual and velocity norms. In the low band $0 < \theta < 1$, point selection can nevertheless fail in the plane: for every prescribed positive proximal scale and clock, a finite, globally Lipschitz, coercive convex objective with a uniform Kurdyka–Łojasiewicz inequality near its minimizer segment admits a same-point initialized trajectory whose exact cluster set is a nondegenerate segment.

Keywords: nonexpansive mapping; fixed point iteration; inertial method; monotone dynamics; weak convergence

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1 Introduction

Let H be a real Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and norm $\|\cdot\|$. A mapping $T : H \to H$ is nonexpansive if

$$\|Tx - Ty\| \le \|x - y\| \qquad (x, y \in H).$$

We assume throughout that $S := \operatorname{Fix} T = \{p \in H : Tp = p\}$ is nonempty. Fixed-point iterations for such mappings provide a common framework for many splitting algorithms; see, for example, [3].

Boţ and Nguyen [6] introduced a Fast Krasnoselskii–Mann method combining weak convergence with a fixed-point residual of order $o(1/k)$. Boţ, Chenchene and Fadili [5] subsequently considered the more general iteration

$$\begin{aligned} x^{k+1} = x^k &+ \frac{\theta}{k+\sigma}(Tx^k - x^k) \\ &+ \left(1 - \frac{\alpha}{k+\sigma}\right)(Tx^k - Tx^{k-1}), \qquad k \geq 0, \end{aligned} \tag{1}$$

where $x^{-1}, x^0 \in H$ are arbitrary. For $\alpha > 2$, $\sigma > 0$ and $1 < \theta < \alpha - 1$, their theorem gives weak convergence and little-$o(1/k)$ decay of the residual and velocity. They also identify the lower endpoint $\theta = 1$ with a Halpern-type iteration. The convergence question at the upper endpoint $\theta = \alpha - 1$ is explicitly left open in [5, Section 6].

Our first result establishes convergence throughout the larger interval $1 < \theta < \alpha$. Together with the known Halpern boundary, a nonlinear lower-band counterexample and scalar upper-band obstructions, it identifies the exact positive-θ range for a universal convergence guarantee. Subsequent results identify the rate change at $\theta = \alpha - 1$. All discrete results concern the unchanged recurrence (1). Related corrected inertial forward–backward methods have a strict parameter margin corresponding to $\theta < \alpha - 1$; see [8, Theorem 2.1]. The history-based forward–backward family in [12, Proposition 5] is another related interior ancestor under its controlled-growth conditions. These results do not give the higher band by a direct parameter substitution. Accelerated $O(1/k)$ fixed-point methods are already known. For example, the recent inertial method of Yang et al. [13] attains this order under a horizon-dependent parameter schedule. Its relation to Fast KM in [13, Appendix A] uses $\theta = \alpha/2$, which lies strictly inside the parameter interval for $\alpha > 2$, rather than the endpoint considered here.

Theorem 1. *Let $T : H \to H$ be nonexpansive with $\operatorname{Fix} T \neq \varnothing$. Suppose that $\alpha > 1$, $\sigma > 0$ and $1 \leq \theta < \alpha$. Then, for every $x^{-1}, x^0 \in H$, the sequence generated by (1) converges weakly to a point $\bar{x} \in \operatorname{Fix} T$. Moreover,*

$$\|x^k - Tx^k\| \longrightarrow 0.$$

In a finite-dimensional Hilbert space, the convergence of the iterates is strong. At $\theta = 1$, strong convergence holds in every Hilbert space, to the projection of the anchor in (3) onto $\operatorname{Fix} T$.

The acceleration boundary $\theta = \alpha - 1$ retains the order $1/k$ of the interior estimates, although the little-o improvement is lost in general.

Theorem 2. *Under the assumptions of Theorem 1, suppose additionally that $\alpha > 2$ and $\theta = \alpha - 1$. Then*

$$\|x^{k+1} - x^k\| = O(1/k), \qquad \|x^k - Tx^k\| = O(1/k).$$

Explicit nonasymptotic bounds are given in Proposition 3.1.

Proposition 7.1 further shows that the two scaled norms have the same finite limit. Proposition 7.2 distinguishes a zero scaled limit from weighted square summability, even for a fixed firmly nonexpansive map.

The order $1/k$ is sharp over the stated class. The following example shows that the little-$o(1/k)$ conclusions valid in the interior cannot be extended to the endpoint. We do not claim optimality of the constants in our upper bounds.

Theorem 3. *Let $H = \mathbb{R}$, $T = -I$, $\alpha = \sigma = 3$, $\theta = 2$ and $x^0 = 1$. The value of x^{-1} is arbitrary. Then*

$$x^{2m} = \frac{1}{2m+1}, \qquad x^{2m+1} = -\frac{1}{2m+3} \qquad (m \geq 0).$$

In particular, $x^k \to 0$, while

$$k|x^k - Tx^k| \longrightarrow 2, \qquad k|x^{k+1} - x^k| \longrightarrow 2.$$

Beyond the acceleration boundary, the iteration still converges but has a slower general rate.

Theorem 4. *Under the assumptions of Theorem 1, suppose that $\max\{1, \alpha - 1\} < \theta < \alpha$ and put $a = \alpha - \theta \in (0, 1)$. Then*

$$\|x^{k+1} - x^k\| = O(k^{-a}), \qquad \|x^k - Tx^k\| = O(k^{-a}).$$

After multiplication by k^a, these two norms have the same finite nonnegative limit. Explicit bounds are given in Proposition 3.2. Fixed reflections attain a positive limit, so the order is sharp; at $\theta = \alpha$, a reflection orbit can fail to converge.

The lower and upper obstructions show that the convergence interval is exact in the following universal sense.

Theorem 5. *Fix $\alpha > 1$ and take $\sigma = \alpha$.*

- *If $0 < \theta < 1$, the fixed nonlinear firmly nonexpansive map on $\ell^2(\mathbb{N}; \mathbb{R}^2)$ and initial pair in (35) have an unbounded orbit, even though $\operatorname{Fix} T = \{0\}$. The same map and initial pair work for each fixed parameter pair in this range.*
- *If $\theta = \alpha$, $T = -I$ on $\mathbb{R}$ and $x^0 = 1$ give $x^k = (-1)^k$. If $\theta > \alpha$, the same choices give $|x^k| \to \infty$.*

Corollary 1.1. *Fix $\alpha > 1$ and $\theta > 0$. The iteration (1) converges weakly to a fixed point for every real Hilbert space, every nonexpansive mapping with a nonempty fixed-point set, every $\sigma > 0$ and every initial pair if and only if*

$$1 \leq \theta < \alpha.$$

The negative cases in Theorem 5 use $\sigma = \alpha$; that theorem does not assert failure for every other fixed clock. Its unbounded-orbit construction below $\theta = 1$ is infinite-dimensional and does not establish finite-dimensional unboundedness. The finite-dimensional proximal nonconvergence result in Theorem 9 below is bounded and covers every prescribed positive clock. The Halpern case is classical [4]; the direct-sum rotation mechanism has standard antecedents [7]. Our contribution there is the controlled nonlinear amplification for this recurrence. The literature comparisons are finite and do not constitute a claim of exhaustive priority.

Within the strictly averaged class, finite dimensionality and boundedness of the fixed-point set instead ensure boundedness and attraction to that set.

Theorem 6. *Let H be finite-dimensional and let $T = (1-\lambda)I + \lambda R$, where $0 < \lambda < 1$ and $R : H \to H$ is nonexpansive. Suppose that $S = \operatorname{Fix} T$ is nonempty and bounded. If $\alpha > 1$, $0 < \theta < 1$ and $\sigma > 0$, then every trajectory of (1) is bounded and satisfies*

$$\operatorname{dist}(x^k, S) \longrightarrow 0, \qquad \|x^k - Tx^k\| \longrightarrow 0.$$

If $S = \{p\}$, then $x^k \to p$ strongly.

The proof first establishes $x^{k+1} - Tx^k = O(k^{-\theta})$ without assuming boundedness. The remaining attraction argument belongs to the established theory of nonexpansive iterations with nonsummable errors [10, 11]; we give its finite-dimensional details in Section 5.4.

Proximal structure also gives residual decay beyond the universal convergence interval. The next statement keeps Hilbert-space residual behavior separate from finite-dimensional iterate convergence.

Theorem 7. *Let H be a real Hilbert space and let $f : H \to \mathbb{R} \cup \{+\infty\}$ be proper, lower semicontinuous and convex, with $\operatorname{argmin} f \neq \varnothing$. Let $T = \operatorname{prox}_{\gamma f}$ with $\gamma > 0$. For every $\alpha > 0$, $\theta > 0$, $\sigma > 0$ and $x^{-1}, x^0 \in H$, the unchanged recurrence (1) satisfies*

$$\|x^k - Tx^k\| \longrightarrow 0, \qquad \|x^{k+1} - x^k\| \longrightarrow 0.$$

If additionally H is finite-dimensional and $\operatorname{argmin} f = \{p\}$, then $x^k \to p$ strongly.

Section 5.5 proves this structural observation using the graph variables and mechanical energy of inertial proximal analysis [1, 2]. No boundedness or weak convergence of the general Hilbert-space trajectory is asserted.

The same two-state energy also applies to a continuous equation. Let $Q : H \to H$ be globally Lipschitz and monotone, and suppose that $S_Q := \{p \in H : Qp = 0\}$ is nonempty. The continuous upper endpoint considered in [5, Section 2.2.1] is

$$\ddot{x}(t) + \frac{\alpha}{t}\dot{x}(t) + \frac{\mathrm{d}}{\mathrm{d}t}Q(x(t)) + \frac{\alpha - 1}{t}Q(x(t)) = 0, \qquad t \geq t_0 > 0. \tag{2}$$

Its general asymptotic behavior is left open in the cited work. We interpret this equation almost everywhere. The solution x is continuously differentiable, its velocity is locally absolutely continuous, and the time derivative of $Q(x(t))$ exists almost everywhere. No differentiability of the mapping Q is assumed.

Theorem 8. *Let $\alpha > 2$, $t_0 > 0$ and $x_0, v_0 \in H$. Then (2), with $x(t_0) = x_0$ and $\dot{x}(t_0) = v_0$, has a unique global solution in the stated class. The trajectory converges weakly to a point of S_Q, and*

$$\|\dot{x}(t)\| = O(1/t), \qquad \|Q(x(t))\| = O(1/t).$$

In finite dimensions the trajectory converges strongly.

Attraction to a compact set of minimizers does not in general imply selection of one minimizer, even for a finite convex objective satisfying a uniform Kurdyka–Łojasiewicz (KL) inequality. The next result identifies this distinction for the proximal low band of the same discrete recurrence. For a bounded sequence, $\omega(x^k)$ denotes the set of all limits of subsequences with indices tending to infinity.

Theorem 9. *Fix $\alpha > 1$, $0 < \theta < 1$, $\gamma > 0$, $\sigma > 0$, and any $p \in (1, 1/\theta]$. There exist a finite, continuous, globally Lipschitz, coercive convex function $\Phi : \mathbb{R}^2 \to \mathbb{R}$ and an initial point $x^0 \in \mathbb{R}^2$ with the following properties:*

$$\min \Phi = 0, \qquad \operatorname{argmin} \Phi = S = [-1, 1] \times \{0\}.$$

For some $C, \delta > 0$, the uniform KL inequality

$$\Phi(x)^{1/p} \le C \operatorname{dist}(0, \partial\Phi(x)) \qquad \big(\operatorname{dist}(x, S) < \delta\big)$$

holds. With $T = \operatorname{prox}_{\gamma\Phi}$ and $x^{-1} = x^0$, the trajectory of (1) is bounded and satisfies

$$\|x^k - Tx^k\| \longrightarrow 0, \qquad \|x^{k+1} - x^k\| \longrightarrow 0, \qquad \Phi(x^k) \longrightarrow 0,$$
$$\omega(x^k) = [-1/\sqrt{2}, 1/\sqrt{2}] \times \{0\}.$$

Thus it neither converges to one point nor has finite total length.

The function and initial point may depend on the prescribed parameters. The theorem concerns an actual trajectory with a same-point initialization for every $\sigma > 0$, including the degenerate first-step case $\sigma = \theta$. Its proof gives an exact low-band drift decomposition, a continuous choice of the initial point, and discrete normal bounds and tangential tracking. A fixed norm infimal convolution preserves every queried proximal graph pair and yields the finite Lipschitz objective. No change of the recurrence or time-dependent objective is used.

Convex inertial nonconvergence under a KL objective is already known in continuous time. Cabot, Engler and Gadat [16, Proposition 4.1] consider one-dimensional flat-bottom convex potentials for

$$\ddot{u}(t) + a(t)\dot{u}(t) + G'(u(t)) = 0.$$

Their condition applies to $a(t) = c/(1+t)$ with $0 < c \le 1$. Taking $G(u) = \operatorname{dist}(u, [-1,1])^2$, which is semialgebraic and satisfies a KL inequality, their proposition together with [16, Corollary 3.1] gives nonstationary trajectories whose cluster set is exactly the minimizer interval. Thus neither convex KL inertial nonconvergence nor an interval cluster set is asserted here as a new general phenomenon. Theorem 9 concerns the particular discrete proximal recurrence (1), its operator-difference correction, all $\alpha > 1$ in the low band, arbitrary prescribed $\gamma, \sigma > 0$, and the specified initialization.

A related geometric precedent is the smooth coercive convex Newton-flow example of Bolte and Pauwels [15, Section 5.6, Corollary 9]. Its identity $\nabla f(x(t)) = e^{-t}\nabla f(x(0))$ implies by convexity that a fixed-direction, vanishing linear-tilt minimizer path need not select one point. That static or continuous mechanism alone does not realize a Fast KM trajectory. The discrete drift, initialization and tracking arguments in Section 8 provide the required link for the present algorithm. Similarly, KL convergence results such as [14, Theorem 2.9] require sufficient-decrease and relative-error conditions in addition to the objective's KL property; the latter is not an algorithm-independent convergence guarantee.

Section 2 proves convergence on the full stated interval. Section 3 proves the rate bounds. Section 4 verifies the reflection examples and upper instability. Section 5 constructs the nonlinear lower-band counterexample and proves the finite-dimensional averaged-map and proximal residual statements. Section 6 treats the continuous endpoint, Section 7 gives scaled-limit and weighted-summability refinements, and Section 8 proves the finite-dimensional proximal nonselection theorem.

2 Proof of Theorem 1

We first dispose of the known case $\theta = 1$. Multiplication of (1) by $k + \sigma$ shows that

$$(k+\sigma)x^{k+1} - (k+\sigma+1-\alpha)Tx^k = (k+\sigma-1)x^k - (k+\sigma-\alpha)Tx^{k-1}.$$

The common vector is $(\alpha - 1)v$, where

$$v = \frac{(\sigma-1)x^0 - (\sigma-\alpha)Tx^{-1}}{\alpha - 1}. \tag{3}$$

Therefore, at every index,

$$x^{k+1} = (1-\lambda_k)Tx^k + \lambda_k v, \qquad \lambda_k = \frac{\alpha-1}{k+\sigma}.$$

On a finite tail, $0 < \lambda_k < 1$, $\lambda_k \to 0$, $\sum_k \lambda_k = \infty$ and $\sum_k |\lambda_{k+1} - \lambda_k| < \infty$. The classical Halpern theorem [4, Theorem 3.1, one-map case], which permits an arbitrary anchor and independent starting point on an unbounded closed convex domain, gives $x^k \to P_S v$ strongly. The finite prefix is harmless. Continuity of $I - T$ also gives residual convergence.

For the remainder of this section, assume $1 < \theta < \alpha$ and put

$$a := \alpha - \theta > 0, \qquad b := \theta - 1 > 0, \qquad A := a + b = \alpha - 1, \qquad t_k := k + \sigma - 1. \tag{4}$$

Choose an integer $K \geq 0$ such that

$$t_K > \max\{1, A\}. \tag{5}$$

The finite prefix needs no sign restriction on its momentum coefficient: $k + \sigma > 0$ and T is defined on all of H.

Proposition 2.1. For $k \geq K$, define

$$z_k := Tx^{k-1}, \qquad w_k := \frac{t_k}{a}x^k - \left(\frac{t_k}{a} - 1\right)Tx^{k-1}, \qquad r_k := z_k - w_k. \tag{6}$$

Then

$$\begin{aligned} x^k &= \left(1 - \frac{a}{t_k}\right) z_k + \frac{a}{t_k} w_k, \\ z_{k+1} &= Tx^k, \\ w_{k+1} &= \left(1 - \frac{b}{t_k}\right) w_k + \frac{b}{t_k} z_k. \end{aligned} \tag{7}$$

For $p \in S$, the energy

$$E_k(p) := b\|z_k - p\|^2 + a\|w_k - p\|^2 \tag{8}$$

satisfies the exact identity

$$\begin{aligned} E_{k+1}(p) - E_k(p) = & -ab\left(\frac{2}{t_k} - \frac{A}{t_k^2}\right)\|r_k\|^2 \\ & - b\big(\|x^k - p\|^2 - \|Tx^k - p\|^2\big). \end{aligned} \tag{9}$$

Consequently, $E_k(p)$ is nonnegative and nonincreasing,

$$\sum_{k=K}^{\infty} \frac{\|r_k\|^2}{t_k} \leq \frac{E_K(p)}{ab}, \tag{10}$$

and all state sequences are bounded. Moreover,

$$x^k - z_k = -\frac{a}{t_k} r_k = O(1/k), \qquad \|r_k\| \leq M := \sqrt{\frac{AE_K(p)}{ab}}. \tag{11}$$

Proof. Write $t = t_k$. Multiplying (1) by $t + 1$ gives

$$(t+1)x^{k+1} = (t + 1 - \theta)x^k + (t + 1 - a)Tx^k - (t + 1 - \alpha)z_k.$$

The coefficient of Tx^k cancels in the definition of w_{k+1}. The remaining terms yield

$$w_{k+1} - w_k = -\frac{\theta - 1}{a}(x^k - z_k) = \frac{b}{t}(z_k - w_k),$$

which proves (7). Its two affine combinations are convex because $t > A > \max\{a, b\}$.

Apply

$$\|(1-c)u + cv\|^2 = (1-c)\|u\|^2 + c\|v\|^2 - c(1-c)\|u - v\|^2$$

to the first and third lines of (7), with weights b and a. The two squared-distance coefficients remain b and a. The coefficient of the subtracted discrepancy square is

$$\frac{ab}{t}\left(1 - \frac{a}{t}\right) + \frac{ab}{t}\left(1 - \frac{b}{t}\right) = ab\left(\frac{2}{t} - \frac{A}{t^2}\right).$$

Replacing x^k by $Tx^k = z_{k+1}$ in the first square gives (9). Nonexpansivity makes its last parenthesis nonnegative, and the discrepancy coefficient is at least ab/t. Telescoping proves (10). Boundedness follows from the positive weights in (8), and the discrepancy bound follows from

$$AE_k(p) - ab\|r_k\|^2 = \|b(z_k - p) + a(w_k - p)\|^2 \geq 0.$$

∎

Proposition 2.2. For every $p \in S$, write $\ell_p := \lim_{k\to\infty} E_k(p)$. Then

$$\|w_k - p\|^2 \longrightarrow \frac{\ell_p}{A}, \quad \|z_k - p\|^2 \longrightarrow \frac{\ell_p}{A}, \quad \|x^k - p\|^2 \longrightarrow \frac{\ell_p}{A}. \tag{12}$$

Proof. For $q_k = \|w_k - p\|^2$, expansion of the last line of (7) gives

$$q_{k+1} = \left(1 - \frac{A}{t_k}\right) q_k + \frac{E_k(p)}{t_k} - v_k, \qquad v_k := \frac{b}{t_k}\left(1 - \frac{b}{t_k}\right)\|r_k\|^2. \tag{13}$$

Here $v_k \geq 0$ and $\sum v_k < \infty$. Deleting the last term gives a scalar averaging recursion, with weights $A/t_k \in (0, 1)$ of divergent sum and targets $E_k(p)/A \to \ell_p/A$. Its values converge to that target: for any $\varepsilon > 0$, the absolute error is eventually bounded by a convex combination of the preceding error and ε, while the products of $1 - A/t_k$ tend to zero.

The difference from this unperturbed recursion tends to zero. Indeed, its contribution from any fixed finite set of v_j tends to zero through those products, while the remaining contribution is bounded by the tail of $\sum v_j$. Thus $q_k \to \ell_p/A$. The other limits follow from $\|z_k - p\|^2 = (E_k(p) - aq_k)/b$ and (11). ∎

Proof of Theorem 1. Subtract consecutive state formulas in (7) to obtain

$$(t_k + 1)(x^{k+1} - x^k) = (t_k + 1 - a)(Tx^k - Tx^{k-1}) + \frac{a(b+1)}{t_k} r_k. \tag{14}$$

With $Y_k = \|x^k - x^{k-1}\|$ and M as in (11), nonexpansivity gives

$$Y_{k+1} \leq \left(1 - \frac{a}{t_k + 1}\right) Y_k + \frac{a}{t_k + 1}\frac{(b+1)M}{t_k}. \tag{15}$$

The weights $a/(t_k+1)$ belong to $(0,1)$ and have divergent sum, and the final targets tend to zero. The same scalar averaging argument therefore gives $Y_k \to 0$. Moreover,

$$\|x^k - Tx^k\| \le \frac{a}{t_k}\|r_k\| + Y_k \longrightarrow 0. \tag{16}$$

Every weak sequential cluster point belongs to S by demiclosedness; the short proof is included. Suppose $x^{k_j} \rightharpoonup x$, set $e_j = x^{k_j} - Tx^{k_j} \to 0$ and $v = Tx - x$. Nonexpansivity gives

$$\|x^{k_j} - x - e_j - v\|^2 \le \|x^{k_j} - x\|^2.$$

Expansion and passage to the limit yield $\|v\|^2 \le 0$, hence $Tx = x$.

Proposition 2.2 gives distance limits to every point of S. If $p, q \in S$ are weak cluster points, the convergent difference

$$\|x^k - p\|^2 - \|x^k - q\|^2 = -2\langle x^k, p - q\rangle + \|p\|^2 - \|q\|^2$$

implies that $\langle x^k, p - q\rangle$ has a limit. Evaluating it along the two weakly convergent subsequences gives $\|p - q\|^2 = 0$. Boundedness and weak sequential compactness now imply weak convergence of the whole sequence. In finite dimensions this is strong convergence. ∎

3 Proofs of Theorems 2 and 4

3.1 The acceleration boundary

Specialize first to $\theta = \alpha - 1$ with $\alpha > 2$. Then $a = 1$ and $b = d := \alpha - 2 > 0$, so $A = d + 1$. Retain the states and energy from Section 2 and fix $p \in S$. We add the scaled velocity and a quadratic correction.

Proposition 3.1. For $k \ge K$, set

$$\begin{aligned} s_k &:= t_k(x^k - x^{k-1}), \qquad c := 2A + \frac{A^2}{d}, \\ G_k(p) &:= \|s_k\|^2 + cE_k(p) - A\|r_k\|^2. \end{aligned} \tag{17}$$

Then

$$G_k(p) = \|s_k\|^2 + 2AE_k(p) + \frac{A}{d}\|d(z_k - p) + (w_k - p)\|^2 \ge 0, \tag{18}$$

and

$$G_{k+1}(p) - G_k(p) \le -A\|r_{k+1} - r_k\|^2 - \frac{A^2(t_k - 1)}{t_k^2}\|r_k\|^2 \le 0. \tag{19}$$

In particular, for every $n \ge K$,

$$\|x^n - x^{n-1}\| \le \frac{\sqrt{G_K(p)}}{t_n}, \tag{20}$$

and

$$\begin{aligned}\|x^n - Tx^n\| &\le \frac{\sqrt{G_K(p)} + \sqrt{AE_K(p)/d}}{t_n} \\ &\le \left(1 + \frac{1}{\sqrt{2d}}\right) \frac{\sqrt{G_K(p)}}{t_n}.\end{aligned} \tag{21}$$

Proof. For vectors $u, v \in H$, one has

$$A\big(d\|u\|^2 + \|v\|^2\big) - d\|u - v\|^2 = \|du + v\|^2.$$

Substitution of $u = z_k - p$ and $v = w_k - p$ proves (18).

To calculate the difference, put $t = t_k$ and $u_k = Tx^k - Tx^{k-1}$. The state and velocity identities (7) and (14) give

$$s_{k+1} = tu_k + \frac{A}{t} r_k, \qquad r_{k+1} = u_k + \left(1 - \frac{d}{t}\right) r_k. \tag{22}$$

Write $F_k = \|s_k\|^2 - A\|r_k\|^2$. Expanding the two squares in (22) yields the exact identity

$$\begin{aligned}F_{k+1} - F_k = &\left(\frac{2Ad}{t} + \frac{A^2}{t^2}\right) \|r_k\|^2 - A\|r_{k+1} - r_k\|^2 \\ &- t^2\big(\|x^k - x^{k-1}\|^2 - \|Tx^k - Tx^{k-1}\|^2\big).\end{aligned} \tag{23}$$

The last parenthesis is nonnegative by nonexpansivity. Proposition 2.1 also gives

$$E_{k+1}(p) - E_k(p) \le -\frac{d}{t}\|r_k\|^2.$$

Since $cd = 2Ad + A^2$, adding c times this estimate to (23) proves

$$G_{k+1}(p) - G_k(p) \le -A\|r_{k+1} - r_k\|^2 - A^2 \left(\frac{1}{t} - \frac{1}{t^2}\right) \|r_k\|^2,$$

which is (19).

It follows from (18) that $\|s_n\| \le \sqrt{G_K(p)}$, proving (20). Furthermore,

$$\|r_n\|^2 \le \frac{A}{d} E_n(p) \le \frac{A}{d} E_K(p), \qquad E_K(p) \le \frac{G_K(p)}{2A}.$$

Combining these inequalities with (16), where $a = 1$, gives (21). ∎

Proof of Theorem 2. Since $t_n = n + \sigma - 1 \sim n$, Proposition 3.1 gives the asserted $O(1/k)$ rates. Applying (20) at $n = k + 1$ gives the velocity indexing in the theorem. ∎

3.2 The higher relaxation band

We now assume $\max\{1, \alpha - 1\} < \theta < \alpha$, so that $0 < a = \alpha - \theta < 1$ and $b = \theta - 1 > 0$, in addition to Theorem 1. The elementary velocity recursion already gives the sharp slower order in this band.

Proposition 3.2. With the notation of Section 2, fix $p \in S$ and define

$$B_K = (t_K+1)^a \|x^K - x^{K-1}\| + a(b+1)M\, 3^a \left(t_K^{a-2} + \frac{t_K^{a-1}}{1-a} \right).$$

For every $n \geq K$,

$$\|x^n - x^{n-1}\| \leq \frac{B_K}{(t_n+1)^a}, \qquad \|x^n - Tx^n\| \leq \frac{aM}{t_n} + \frac{B_K}{(t_n+1)^a}. \tag{24}$$

There is a finite $L \geq 0$ such that

$$k^a \|x^k - Tx^k\| \longrightarrow L, \qquad k^a \|x^{k+1} - x^k\| \longrightarrow L. \tag{25}$$

Proof. Let $D = a(b+1)M$ and $P_k = 1 - a/(t_k+1)$. For $l < n$,

$$\prod_{k=l}^{n-1} P_k \leq \exp\left(-a \sum_{k=l}^{n-1} \frac{1}{t_k+1} \right) \leq \left(\frac{t_l+1}{t_n+1} \right)^a,$$

where the harmonic sum dominates the integral from t_l+1 to t_n+1. Iterating (15) gives

$$(t_n+1)^a Y_n \leq (t_K+1)^a Y_K + D \sum_{j=K}^{n-1} \frac{(t_j+2)^a}{t_j(t_j+1)}.$$

Since $t_j > 1$, the summands are at most $3^a t_j^{a-2}$. Their sum is bounded by

$$3^a \left(t_K^{a-2} + \int_{t_K}^{\infty} u^{a-2}\, \mathrm{d}u \right) = 3^a \left(t_K^{a-2} + \frac{t_K^{a-1}}{1-a} \right).$$

This proves the velocity bound, and (16) proves the residual bound.

For the norm limit, put

$$\nu_k = \frac{\Gamma(t_k+1)}{\Gamma(t_k+1-a)}.$$

All gamma arguments are positive on the chosen tail. The functional equation gives $\nu_{k+1} P_k = \nu_k$, and the gamma-ratio asymptotic gives $\nu_k \sim t_k^a$; see [9, Section 5.11]. Thus

$$\nu_{k+1} Y_{k+1} \leq \nu_k Y_k + \varepsilon_k, \qquad \varepsilon_k = \frac{D\nu_{k+1}}{t_k(t_k+1)}, \qquad \sum_{k=K}^{\infty} \varepsilon_k < \infty.$$

The sequence $\nu_k Y_k + \sum_{j=k}^{\infty} \varepsilon_j$ is nonnegative and nonincreasing. Therefore $k^a Y_k$ has a finite limit. Finally, (14) gives the exact identity

$$x^k - Tx^k = -\frac{t_k+1}{t_k+1-a}(x^{k+1} - x^k) - \frac{a(t_k - A)}{t_k(t_k+1-a)} r_k.$$

The last term is $O(1/k)$, which is negligible on the k^a scale. This proves (25) with the stated forward-velocity indexing. ∎

Proof of Theorem 4. The bounds and limits follow from Proposition 3.2 and $t_k \sim k$. Their sharpness and the upper-boundary obstruction are verified in Proposition 4.1 below. ∎

4 Sharpness and the upper convergence boundary

4.1 The acceleration boundary reflection

Proof. For the stated parameters, (1) reduces to

$$(k+3)x^{k+1} = -x^k + kx^{k-1}, \qquad k \geq 0. \tag{26}$$

The initial step is $x^1 = -1/3$ and is independent of x^{-1}. Together with $x^0 = 1$, this verifies the formulas at $m = 0$.

Assume the formulas hold at the indices needed in (26). If $k = 2m \geq 2$, then

$$(2m+3)x^{2m+1} = -\frac{1}{2m+1} - \frac{2m}{2m+1} = -1.$$

If $k = 2m+1$, then

$$(2m+4)x^{2m+2} = \frac{1}{2m+3} + \frac{2m+1}{2m+1} = \frac{2m+4}{2m+3}.$$

Induction proves the orbit formulas for every $m \geq 0$. They imply $x^k \to 0$. Since $|x^k - Tx^k| = 2|x^k|$, the residual limit follows along both parities. Finally,

$$|x^{2m+1} - x^{2m}| = \frac{1}{2m+3} + \frac{1}{2m+1}, \qquad |x^{2m+2} - x^{2m+1}| = \frac{2}{2m+3}.$$

Multiplication by $2m$ and $2m+1$, respectively, gives the velocity limit. ∎

4.2 The higher band and the upper boundary

Proposition 4.1. Suppose $\alpha > 1$, $\max\{1, \alpha - 1\} < \theta < \alpha$ and $a = \alpha - \theta$. Set $\sigma = \alpha$, $T = -I$ on $\mathbb{R}$ and $x^0 = 1$, with arbitrary x^{-1}. Then

$$x^k = \frac{1}{B(\theta, a)} \int_0^1 (1-2u)^k u^{\theta-1}(1-u)^{a-1}\,\mathrm{d}u, \tag{27}$$

and

$$(-1)^k k^a x^k \longrightarrow \frac{\Gamma(\alpha)}{2^a \Gamma(\theta)}. \tag{28}$$

Both limits in (25) equal $2^{1-a}\Gamma(\alpha)/\Gamma(\theta) > 0$ for this orbit. At $\theta = \alpha$, the same map, initialization and choice $\sigma = \alpha$ instead give $x^k = (-1)^k$, so general weak convergence fails at that boundary.

Proof. The reflection recurrence is

$$(k+\alpha)x^{k+1} = (\alpha - 2\theta)x^k + kx^{k-1}, \qquad x^1 = \frac{\alpha - 2\theta}{\alpha}.$$

The initial step is independent of x^{-1}. Let M_k denote the right side of (27). The beta identity $B(\theta, a) = \Gamma(\theta)\Gamma(a)/\Gamma(\alpha)$ gives $M_0 = 1$ and $M_1 = (a-\theta)/\alpha$; see [9, Section 5.12]. For $k \geq 1$, integrating the derivative of

$$u^\theta (1-u)^a (1-2u)^k$$

over $[0,1]$ is legitimate since $a,\theta > 0$ and both endpoint values vanish. On writing $Y = 1-2u$, the resulting identity is

$$0 = (\theta - a)M_k + \alpha M_{k+1} - k(M_{k-1} - M_{k+1}).$$

Thus the moments satisfy the exact recurrence, proving (27).

To prove the asymptotic, split the integral at $u = 1/2$. Its left part is $O(k^{-\theta})$, since $(1-2u)^k \le e^{-2ku}$ there and $(1-u)^{a-1} \le 2^{1-a}$. On the right, set $s = k(1-u)$. After multiplication by $(-1)^k k^a$, this part becomes

$$\int_0^{k/2} \left(1 - \frac{2s}{k}\right)^k \left(1 - \frac{s}{k}\right)^{\theta-1} s^{a-1}\,\mathrm{d}s.$$

Extend the integrand by zero past $k/2$. It is bounded by the integrable function $e^{-2s}s^{a-1}$ and converges to it pointwise. Dominated convergence gives the limit $\Gamma(a)/2^a$. The left part vanishes on this scale because $\theta > 1 > a$. Division by $B(\theta,a)$ proves (28). The residual is $2|x^k|$, and consecutive leading terms alternate in sign, so the scaled velocity has the same positive limit.

At $\theta = \alpha$, use the original recurrence directly, without the singular $a = 0$ transformation. It becomes

$$(k+\alpha)x^{k+1} = -\alpha x^k + kx^{k-1}.$$

Its initial step gives $x^1 = -1$, and induction gives $x^k = (-1)^k$ for all k. ∎

For $\theta > \alpha$, set $c = \theta - \alpha > 0$ and $y_k = (-1)^k x^k$ in the same reflection recurrence. Then

$$(k+\alpha)y_{k+1} = (\alpha + 2c)y_k + ky_{k-1}, \qquad y_0 = 1, \quad y_1 = 1 + 2c/\alpha.$$

All y_k are positive. If $m_k = \min\{y_{k-1}, y_k\}$ for $k \ge 1$, then

$$y_{k+1} \ge \left(1 + \frac{2c}{k+\alpha}\right) m_k, \qquad m_{k+1} \ge m_k.$$

Applying the first inequality once more gives

$$m_{k+2} \ge \left(1 + \frac{2c}{k+1+\alpha}\right) m_k.$$

The product on either parity diverges because the corresponding harmonic series diverges. Hence $m_k \to \infty$ and $|x^k| \to \infty$, completing the upper part of Theorem 5.

5 An unbounded orbit below the Halpern boundary

We prove the first part of Theorem 5. The direct-sum use of small-angle rotations is a standard mechanism; compare [7, Example 1 and Theorems 8–9]. Here each block also contains a halfspace projection, and averaging with the identity makes the final map firmly nonexpansive. The main point is to control a long affine branch of the unchanged nonlinear iteration, so that its finite amplification can be combined into one unbounded orbit.

5.1 A projection–rotation block

Fix $\alpha > 1$, $0 < \theta < 1$, and set

$$a = \alpha - \theta > 0, \qquad h = \frac{1}{2}, \qquad q = \frac{3+\theta}{4(1+\theta)}.$$

Then $1/2 < q < 1/(1+\theta) < 1$. For $0 < \varepsilon < 1/4$, put

$$c = 1-\varepsilon, \quad s = \sqrt{1-c^2}, \quad R_\varepsilon = \begin{pmatrix} c & -s \\ s & c \end{pmatrix}, \quad C = \{v \in \mathbb{R}^2 : v_1 \le h\}, \quad T_\varepsilon = \frac{I + R_\varepsilon P_C}{2}.$$

The map $R_\varepsilon P_C$ is nonexpansive and fixes zero; its average with the identity is firmly nonexpansive. These two maps have the same fixed points. If $v_1 > h$, then $\|P_C v\| < \|v\|$, which excludes $R_\varepsilon P_C v = v$; if $v_1 \le h$, fixedness implies $R_\varepsilon v = v$ and hence $v = 0$. Thus $\operatorname{Fix} T_\varepsilon = \{0\}$.

On the active halfspace $v_1 > h$, the map equals $Mv + v_*$, where

$$M = \begin{pmatrix} 1/2 & -s/2 \\ 0 & (1+c)/2 \end{pmatrix}, \qquad v_* = \frac{h}{2}\begin{pmatrix} c \\ s \end{pmatrix}.$$

Its affine extension has fixed point $\xi_* = (-h, hs/(1-c))$, outside the active halfspace. This point is only an algebraic device, not a fixed point of T_ε.

Let $\xi^0 = (1,0)$ and $\xi^{-1} = 0$, with $\sigma = \alpha$. Define

$$p_k(z) = \frac{1}{B(\theta,a)} \int_0^1 (1-u+uz)^k u^{\theta-1}(1-u)^{a-1}\,\mathrm{d}u. \tag{29}$$

For $k \ge 1$, integration of the derivative of $u^\theta(1-u)^a(1-u+uz)^k$ gives

$$(k+\alpha)p_{k+1}(z) = [(k+a) + (k+\theta)z]p_k(z) - kzp_{k-1}(z). \tag{30}$$

Indeed, with $f = 1-u+uz$, the polynomial multiplying f^{k-1} in the difference between the two sides equals

$$(1-z)\big[(\theta - \alpha u)f - k(1-z)u(1-u)\big].$$

Both boundary values vanish. Also $p_0 = 1$ and $p_1 = (a+\theta z)/\alpha$.

Translation by ξ_* therefore gives the affine candidate orbit $\xi_* + p_k(M)(\xi^0 - \xi_*)$. The distinct eigenvalues are $\beta = 1/2$ and $\zeta = (1+c)/2 = 1 - \varepsilon/2$, so

$$p_k(M) = p_k(\beta)I + \frac{p_k(\zeta) - p_k(\beta)}{\zeta - \beta}(M - \beta I).$$

Since $(M - \beta I)/(\zeta - \beta) = \left(\begin{smallmatrix} 0 & -s/c \\ 0 & 1 \end{smallmatrix}\right)$, its coordinates are

$$\begin{aligned} X_k &= -h + \frac{h(1+c)}{c}p_k(\zeta) + \left(1 - \frac{h}{c}\right)p_k(\beta), \\ Y_k &= \frac{hs}{1-c}\big(1 - p_k(\zeta)\big). \end{aligned} \tag{31}$$

We next prove that these formulas describe the actual nonlinear orbit throughout the required interval.

5.2 A uniform active-branch interval

Let U have beta density with parameters (θ, a), so that $\mathbb{E}U = \theta/\alpha$. Write $D_k = 1 - p_k(\zeta)$. Bernoulli's inequality gives

$$0 \le D_k = \mathbb{E}[1 - (1 - \varepsilon U/2)^k] \le \frac{\theta}{2\alpha} k\varepsilon. \tag{32}$$

Moreover,

$$p_k(\beta) \ge p_k(0) = \prod_{l=0}^{k-1} \left(1 - \frac{\theta}{\alpha + l}\right) \ge C_0 (k + a)^{-\theta}, \qquad C_0 = a^\theta e^{-\theta/a} > 0.$$

For $k \ge 1$, this follows from $\log(1 - v) \ge -v/(1 - v)$ and

$$\sum_{l=0}^{k-1} \frac{1}{a + l} \le \frac{1}{a} + \log \frac{k + a}{a};$$

the bound is also valid at $k = 0$.

Rewriting the first line of (31), and using $c \ge 3/4$, yields

$$\begin{aligned} X_k - h &= \frac{h(1 - c)}{c} + \left(1 - \frac{h}{c}\right) p_k(\beta) - \frac{h(1 + c)}{c} D_k \\ &\ge \frac{C_0}{3} (k + a)^{-\theta} - \frac{2\theta}{\alpha} k\varepsilon. \end{aligned} \tag{33}$$

Let $N_\varepsilon = \lfloor \varepsilon^{-q} \rfloor$ and $\delta = (1 - \theta)/4 > 0$. Since $1 - q(1 + \theta) = \delta$, for $0 < \varepsilon < 1$,

$$\varepsilon N_\varepsilon (N_\varepsilon + a)^\theta \le (1 + a)^\theta \varepsilon^\delta \longrightarrow 0.$$

For example, take

$$\varepsilon_0 = \min \left\{ \frac{1}{4},\ 2^{-1/q},\ \left(\frac{\alpha C_0}{12\theta (1 + a)^\theta} \right)^{1/\delta} \right\}.$$

If $0 < \varepsilon < \varepsilon_0$, inequality (33) is strictly positive simultaneously for every $0 \le k \le N_\varepsilon$. Induction now identifies the actual orbit with (31) on that interval. Although ξ^{-1} is not on the active face, its coefficient in the initial step is zero because $\sigma = \alpha$. Each subsequent queried point in the induction is on the verified affine branch.

For $k\varepsilon \le 1$ and $0 \le u \le 1$,

$$1 - (1 - \varepsilon u/2)^k \ge \frac{k\varepsilon u/2}{1 + k\varepsilon u/2} \ge \frac{1}{4} k\varepsilon u.$$

The first inequality follows by applying Bernoulli's inequality to $(1 - v)^{-k}$ with $v = \varepsilon u/2$. Thus $D_k \ge \theta k\varepsilon/(4\alpha)$. For the above range of ε, we have $N_\varepsilon \varepsilon \le 1$ and $N_\varepsilon \ge \varepsilon^{-q}/2$. Since $s/(1 - c) \ge \varepsilon^{-1/2}$, the second line of (31) gives

$$Y_{N_\varepsilon} \ge \frac{\theta}{16\alpha} \varepsilon^{1/2 - q}. \tag{34}$$

This proves arbitrarily large finite transients, without asserting unboundedness for any one two-dimensional block.

5.3 One fixed Hilbert-space counterexample

Take $H = \ell^2(\mathbb{N};\mathbb{R}^2)$, with $j \geq 1$, and define once and for all

$$\begin{aligned} \rho_j &= 2^{-j}, \qquad \varepsilon_j = e^{-j^2-2}, \qquad c_j = 1-\varepsilon_j, \quad s_j = \sqrt{1-c_j^2}, \\ R_j &= \begin{pmatrix} c_j & -s_j \\ s_j & c_j \end{pmatrix}, \qquad C_j = \{v \in \mathbb{R}^2 : v_1 \leq \rho_j/2\}, \\ (Tx)_j &= \frac{x_j + R_j P_{C_j}(x_j)}{2}, \qquad x_j^0 = (\rho_j, 0), \quad x_j^{-1} = 0. \end{aligned} \tag{35}$$

Let $(Sx)_j = R_j P_{C_j}(x_j)$. Each block of S is nonexpansive, has unique fixed point zero, and has norm at most the input norm. Hence $S : H \to H$ is well-defined and nonexpansive. Since $T = (I+S)/2$,

$$\langle Tx - Ty, x - y\rangle - \|Tx - Ty\|^2 = \frac{1}{4}\big(\|x-y\|^2 - \|Sx - Sy\|^2\big) \geq 0.$$

Thus T is firmly nonexpansive and $\operatorname{Fix} T = \operatorname{Fix} S = \{0\}$. The initial norm is $\|x^0\|^2 = \sum_{j\geq 1} 4^{-j} = 1/3$. Every finite iterate belongs to H because the recurrence combines elements of H and their images under T.

The scaled halfspaces give the exact identity

$$\frac{\rho_j v + R_j P_{C_j}(\rho_j v)}{2} = \rho_j T_{\varepsilon_j}(v).$$

It follows by induction that $x_j^k = \rho_j \xi^k(\varepsilon_j)$ for every finite k. Now fix any $\alpha > 1$ and $0 < \theta < 1$, and let $N_j = \lfloor \varepsilon_j^{-q} \rfloor$. All sufficiently large j satisfy the uniform branch threshold above, so

$$\|x^{N_j}\| \geq |x_{j,2}^{N_j}| \geq \frac{\theta}{16\alpha}\, 2^{-j} e^{(q-1/2)(j^2+2)} \longrightarrow \infty.$$

The map and initialization in (35) do not depend on α or θ. Only the observation subsequence and its starting index depend on those parameters. Thus this is one unbounded orbit for each fixed parameter pair, and not a sequence of maps chosen at successive horizons. Weak convergence would imply boundedness, so it fails.

Remark. The obstruction uses nonlinearity. If T is a bounded linear contraction and $\sigma = \alpha$ with $0 < \theta < \alpha$, the same polynomial identity gives

$$x^k = p_k(T)x^0 = \frac{1}{B(\theta, \alpha-\theta)} \int_0^1 ((1-u)I + uT)^k x^0\, u^{\theta-1}(1-u)^{\alpha-\theta-1}\, \mathrm{d}u.$$

Each operator inside the integral is a contraction. Consequently $\|x^k\| \leq \|x^0\|$ for every k, so this linear subclass cannot display the amplification used above.

5.4 Finite dimensional averaged maps

The preceding counterexample has a bounded singleton fixed-point set and is firmly nonexpansive, but its space is infinite-dimensional. We now prove Theorem 6. The algorithm-specific step is a comparison estimate for the error in a Picard iteration. The subsequent vanishing-error attraction is consistent with the established perturbation theory of Reich and Zaslavski [10, 11]; we include the elementary finite-dimensional argument to make the boundedness requirement explicit.

Proposition 5.1. Let $T : H \to H$ be nonexpansive with $p \in \operatorname{Fix} T$, and suppose that $\alpha > 1$, $0 < \theta < 1$ and $\sigma > 0$. Put

$$a = \alpha - \theta > 0, \qquad c = 1 - \theta > 0, \qquad t_k = k + \sigma - 1,$$

and choose K such that $t_K > \max\{1, a, c\}$. Define z_k, w_k, r_k by (6) and set

$$J_k(p) = c\|z_k - p\|^2 + a\|w_k - p\|^2.$$

Then, for $k \geq K$,

$$J_{k+1}(p) \leq \left(1 + \frac{2c}{t_k} + \frac{(a+c)^2}{t_k^2}\right) J_k(p). \tag{36}$$

Consequently,

$$J_k(p) = O(t_k^{2c}), \qquad \|x^{k+1} - Tx^k\| = O(k^{-\theta}) \longrightarrow 0. \tag{37}$$

No boundedness assumption on the trajectory or finite-dimensional assumption is used in this proposition.

Proof. The same cancellation that gives (7) now yields

$$x^k = \left(1 - \frac{a}{t_k}\right) z_k + \frac{a}{t_k} w_k, \qquad z_{k+1} = Tx^k, \qquad w_{k+1} = \left(1 + \frac{c}{t_k}\right) w_k - \frac{c}{t_k} z_k.$$

Unlike the update in the convergence band, the last combination is not convex. Direct expansion, with $t = t_k$, gives

$$\begin{aligned} J_{k+1}(p) - J_k(p) = \frac{2ac}{t}\big(\|w_k - p\|^2 - \|z_k - p\|^2\big) + \frac{ac(a+c)}{t^2}\|r_k\|^2 \\ - c\big(\|x^k - p\|^2 - \|Tx^k - p\|^2\big). \end{aligned} \tag{38}$$

The last parenthesis is nonnegative. Also

$$ac\|r_k\|^2 \leq (a+c) J_k(p), \qquad a\|w_k - p\|^2 \leq J_k(p),$$

so (36) follows. Its product is $O(t_k^{2c})$, because the logarithm of each factor is at most $2c/t_k + (a+c)^2/t_k^2$. Finally,

$$x^{k+1} - Tx^k = -\frac{a}{t_{k+1}} r_{k+1}, \qquad \|x^{k+1} - Tx^k\|^2 \leq \frac{a(a+c)}{c\, t_{k+1}^2} J_{k+1}(p).$$

Since $c - 1 = -\theta$, this proves (37). The error controlled here is $x^{k+1} - Tx^k$, not yet the fixed-point residual $x^k - Tx^k$. ∎

Proof of Theorem 6. The case $H = \{0\}$ is immediate. Assume $\dim H \geq 1$ and write $\nu = (1 - \lambda)/\lambda > 0$. The averaged representation gives, for every $p \in S$,

$$\|Tx - p\|^2 \leq \|x - p\|^2 - \nu\|x - Tx\|^2. \tag{39}$$

We first record a radial decrease that follows from compactness and nonexpansivity. Translate one $p \in S$ to zero, and choose $R_0 > 0$ such that S lies strictly inside the ball of radius R_0. On its boundary, (39) gives

$$R_0 - \|T(R_0 u)\| > 0 \qquad (\|u\| = 1).$$

The unit sphere is compact, so the left side has a positive minimum δ. For $t \geq R_0$, nonexpansivity implies

$$\|T(tu)\| \leq \|T(R_0 u)\| + t - R_0 \leq t - \delta.$$

By Proposition 5.1, the actual trajectory satisfies $x^{k+1} = Tx^k + e_k$ with $e_k \to 0$. Eventually $\|e_k\| < \delta/2$. Its radius therefore decreases by at least $\delta/2$ whenever $\|x^k\| \geq R_0$; from inside that ball the next radius is at most $R_0 + \delta/2$. Thus the trajectory is bounded.

For completeness, attraction also follows by a compactness argument. Write $d(x) = \operatorname{dist}(x, S)$. Taking a nearest fixed point in (39) gives

$$d(Tx)^2 \leq d(x)^2 - \nu\|x - Tx\|^2.$$

Hence $d(Tx) < d(x)$ off S. On a closed ball containing the trajectory, the continuous function $d(x) - d(Tx)$ has a positive minimum η_ε on the compact set where $d(x) \geq \varepsilon/2$, if that set is nonempty. Once $\|e_k\| < \min\{\eta_\varepsilon/2, \varepsilon/2\}$, the distance decreases by at least $\eta_\varepsilon/2$ whenever it is at least $\varepsilon/2$. It must enter the smaller neighborhood in finitely many steps and remains within distance ε thereafter. Thus $d(x^k) \to 0$. Finally,

$$\|x^k - Tx^k\| \leq 2d(x^k) \longrightarrow 0.$$

If $S = \{p\}$, distance convergence is strong convergence to p. ∎

The perturbation step is a special case of established nonsummable-error results: [10, Theorems 2.1–2.2] treats sufficiently small errors under bounded-set-uniform attraction, and [11, Theorem 2.1] gives distance convergence for a bounded inexact orbit with vanishing errors. In the present finite-dimensional averaged setting, exact Picard iterates approach S uniformly on every compact initial ball. Indeed, (39) makes the exact orbit bounded with summable residual squares, so compactness and monotonicity of its distance give pointwise attraction. These distance functions are continuous and nonincreasing in the iteration index; compactness then gives uniform convergence. The radial argument

supplies the required boundedness of the inexact orbit. Neither these general robustness principles nor the compact-sphere observation is claimed as new. The comparison (37) connects them to the unchanged Fast KM recurrence.

In particular, each fixed two-dimensional block T_ε used above converges to zero under these parameters, despite the arbitrarily large transients across the block family. The single infinite direct sum in (35) has an unbounded orbit. The finite-dimensional conclusion requires bounded S and does not assert convergence to one selected point when S contains several points. Theorem 9 shows that this distinction cannot be removed even for finite Lipschitz coercive convex proximal objectives with a uniform KL inequality.

5.5 Proximal residual decay for positive parameters

We prove Theorem 7. The graph-variable formulation and mechanical-energy argument belong to the established analysis of inertial proximal algorithms with Hessian-driven damping; see [1, 2]. We use them here to identify the residual behavior of the unchanged recurrence for positive parameters, without asserting a new proximal method.

Let $f_* := \min f$, and define

$$z_k = Tx^{k-1}, \qquad g_k = \frac{x^{k-1} - z_k}{\gamma} \in \partial f(z_k) \quad (k \geq 0), \qquad v_k = z_k - z_{k-1} \quad (k \geq 1).$$

Every z_k belongs to $\operatorname{dom} f$, even if an initial point does not, and f_* is finite. Choose $K \geq 0$ with $K + \sigma > \max\{\alpha, \theta\}$. For $k \geq K$, write $n = k + \sigma$ and set

$$\mathcal{E}_k = \gamma\theta\big(f(z_{k+1}) - f_*\big) + \frac{n-\alpha}{2}\|v_{k+1}\|^2. \tag{40}$$

These quantities are finite and nonnegative. No function value at x^k is used.

Proof of Theorem 7. The original recurrence gives the exact graph relation

$$nv_{k+2} - (n-\alpha)v_{k+1} + \gamma(n-\theta)(g_{k+2} - g_{k+1}) + \gamma\theta g_{k+2} = 0. \tag{41}$$

Put

$$D_k = \langle g_{k+2}, v_{k+2}\rangle - \big(f(z_{k+2}) - f(z_{k+1})\big) \geq 0.$$

Taking the inner product of (41) with v_{k+2} and expanding the kinetic difference yields

$$\begin{aligned} \mathcal{E}_{k+1} - \mathcal{E}_k = & -\left(\alpha - \frac{1}{2}\right)\|v_{k+2}\|^2 - \frac{n-\alpha}{2}\|v_{k+2} - v_{k+1}\|^2 \\ & - \gamma(n-\theta)\langle g_{k+2} - g_{k+1}, v_{k+2}\rangle - \gamma\theta D_k. \end{aligned} \tag{42}$$

Subgradient monotonicity makes the inner product nonnegative. If $\alpha > 1/2$, telescoping gives

$$\left(\alpha - \frac{1}{2}\right)\sum_{k=K}^{\infty}\|v_{k+2}\|^2 \leq \mathcal{E}_K,$$

so $v_k \to 0$.

For $0 < \alpha \le 1/2$, use $\mathcal{E}_{k+1} \ge (n+1-\alpha)\|v_{k+2}\|^2/2$ in the same identity. This gives

$$\mathcal{E}_{k+1} \le \frac{n+1-\alpha}{n+\alpha}\mathcal{E}_k.$$

The ratio is $1 + (1-2\alpha)/(n+\alpha)$. Taking products therefore yields

$$\mathcal{E}_k = O((k+\sigma)^{1-2\alpha}), \qquad \|v_{k+1}\| = O((k+\sigma)^{-\alpha}) \longrightarrow 0. \tag{43}$$

When $\alpha < 1/2$, this argument permits growth of the energy. It gives neither a fixed objective-sublevel bound nor square summability of v_k.

To recover the actual residual, put $q_k = \gamma g_{k+1} + v_{k+1}$. Equation (41) is equivalent to

$$q_{k+1} = \left(1 - \frac{\theta}{n}\right) q_k + \frac{\theta}{n}\left(\frac{\theta-\alpha}{\theta} v_{k+1}\right).$$

On the chosen tail, the averaging weights lie in $(0,1)$ and have divergent sum, while the target vectors tend to zero. Given $\varepsilon > 0$, their norms are eventually at most ε; iteration bounds $\|q_k\|$ by ε plus a fixed initial term multiplied by a product of $1 - \theta/n$ tending to zero. Hence $q_k \to 0$, and

$$x^k - Tx^k = \gamma g_{k+1} = q_k - v_{k+1} \longrightarrow 0.$$

The original recurrence now gives

$$x^{k+1} - x^k = -\frac{\theta}{n}\gamma g_{k+1} + \left(1 - \frac{\alpha}{n}\right) v_{k+1} \longrightarrow 0.$$

No compactness or boundedness of the trajectory was used in these Hilbert-space conclusions.

For the finite-dimensional assertion, we use an elementary fact about approximate fixed points. If a nonexpansive map on $\mathbb{R}^d$ has a nonempty bounded fixed-point set, every sequence u_j with $u_j - Tu_j \to 0$ is bounded. To prove this, translate a fixed point to zero and suppose that a subsequence has $r_j = \|u_j\| \to \infty$ and $u_j/r_j \to d$, with $\|d\| = 1$. Put $e_j = u_j - Tu_j \to 0$. For every fixed y, nonexpansivity gives

$$\|Ty - u_j + e_j\|^2 \le \|y - u_j\|^2.$$

Expansion and division by $2r_j$ show that $\langle d, Ty - y\rangle \ge 0$. Taking $y = td$ gives $\langle d, T(td)\rangle \ge t$, whereas $\|T(td)\| \le t$. Thus $T(td) = td$ for every $t \ge 0$, contradicting boundedness of the fixed-point set. This proves the fact.

For $T = \operatorname{prox}_{\gamma f}$, $\operatorname{Fix} T = \operatorname{argmin} f$. When this set is $\{p\}$, residual convergence and the preceding fact give boundedness of x^k. Every cluster point is fixed by continuity of T, so finite-dimensional compactness implies $x^k \to p$. ∎

The relation to the smooth IPAHD family is concrete. On the positive shifted tail clock $q = n - \alpha$, its step-one coefficients can be chosen as

$$\beta_q = \frac{\gamma(q + \alpha - \theta)}{q}, \qquad b_q = \frac{\gamma\theta}{q}.$$

They reproduce (41), while the quantity in the hypotheses of [2, Theorem 3.1] is

$$B_q = q(b_q + \beta_q - \beta_{q+1}) - \beta_{q+1} = \gamma(\theta - 1).$$

The printed assumptions $\alpha > 1$ and $B_q > 0$ therefore exclude small α and $\theta \leq 1$. The additional ratio condition in its iterate-convergence theorem is also not met by this substitution: for $\theta > 1$, $\beta_{q+1}/B_q \to 1/(\theta - 1)$. This comparison concerns the author manuscript compiled 11 October 2021, linked in the bibliography; it does not claim that no other known argument implies the present residual result.

The conclusion of Theorem 7 is restricted to residual and step decay in general Hilbert space. It does not establish boundedness or weak convergence of the full trajectory there. In finite dimensions, Theorem 9 shows that convergence among multiple minimizers cannot be added as a general conclusion in the low band, even under its stronger function-class assumptions. In particular, no inconsistency with either the infinite-dimensional unbounded-orbit result or that bounded nonselection example is inferred from residual decay alone.

6 Proof of Theorem 8

Proof. Write $d = \alpha - 2$ and $A = d + 1$. Define

$$w = x + t(\dot{x} + Q(x)), \qquad r = x - w. \tag{44}$$

The equation is equivalent to

$$\dot{x} = -Q(x) + \frac{w - x}{t}, \qquad \dot{w} = \frac{d}{t}(x - w), \tag{45}$$

with $w(t_0) = x_0 + t_0(v_0 + Q(x_0))$. Indeed, differentiation of (44) and substitution in (2) give $\dot{w} = -d(\dot{x} + Q(x))$; the converse follows by differentiating the first equation in (45). The vector field in (45) is globally Lipschitz in the state on each bounded time interval and has at most linear growth. The Hilbert-space Picard theorem and the usual growth estimate therefore give a unique global C^1 solution. On each compact time interval, $Q(x(\cdot))$ and $\dot{x}$ are Lipschitz. Hilbert-valued Lipschitz curves have strong derivatives almost everywhere, which gives the stated solution regularity. All energies below are locally absolutely continuous.

Fix $p \in S_Q$ and put

$$E_p(t) = d\|x(t) - p\|^2 + \|w(t) - p\|^2.$$

Direct differentiation yields

$$\dot{E}_p = -2d\langle x - p, Q(x)\rangle - \frac{2d}{t}\|r\|^2 \le 0. \tag{46}$$

Thus x, w, r are bounded and

$$\int_{t_0}^{\infty} \frac{\|r(t)\|^2}{t}\,\mathrm{d}t \le \frac{E_p(t_0)}{2d}. \tag{47}$$

The stronger rate follows from the same correction used for the discrete iteration. Set $c = 2A + A^2/d$ and

$$G_p(t) = t^2\|\dot{x}(t)\|^2 + cE_p(t) - A\|r(t)\|^2. \tag{48}$$

The identity

$$G_p = t^2\|\dot{x}\|^2 + 2AE_p + \frac{A}{d}\|d(x-p) + (w-p)\|^2$$

shows that $G_p \ge 0$. If $s = t\dot{x}$, equations (45) give

$$\dot{s} = -t\frac{\mathrm{d}}{\mathrm{d}t}Q(x) + \frac{A}{t}r, \qquad \dot{r} = \frac{1}{t}(s - dr).$$

Consequently, for almost every $t \ge t_0$,

$$\dot{G}_p = -2t^2\langle \dot{x}, \frac{\mathrm{d}}{\mathrm{d}t}Q(x)\rangle - 2cd\langle x - p, Q(x)\rangle - \frac{2d(c-A)}{t}\|r\|^2 \le 0. \tag{49}$$

The first inner product is nonnegative: apply monotonicity to $x(t+h), x(t)$, divide by h^2, and pass to the time derivatives at a point where they exist. This uses no derivative of Q as an operator. The second inner product is nonnegative because $Qp = 0$.

It follows that

$$\|\dot{x}(t)\| \le \frac{\sqrt{G_p(t_0)}}{t}, \qquad \|Q(x(t))\| \le \frac{\sqrt{G_p(t_0)} + \sqrt{AE_p(t_0)/d}}{t}. \tag{50}$$

For the second estimate, use $Q(x) = -\dot{x} - r/t$ and $\|r\|^2 \le AE_p/d$.

It remains to prove convergence of the trajectory. Let $q(t) = \|w(t) - p\|^2$ and $\ell_p = \lim_{t\to\infty} E_p(t)$. The exact scalar identity is

$$\dot{q} + \frac{A}{t}q = \frac{E_p(t)}{t} - \frac{d}{t}\|r(t)\|^2.$$

Multiplication by t^A and integration give

$$q(t) = \left(\frac{t_0}{t}\right)^A q(t_0) + \frac{1}{t^A}\int_{t_0}^{t} u^{A-1}E_p(u)\,\mathrm{d}u - \int_{t_0}^{t}\left(\frac{u}{t}\right)^A \frac{d\|r(u)\|^2}{u}\,\mathrm{d}u.$$

The middle term tends to ℓ_p/A: replace $E_p(u)$ by ℓ_p and bound the error after a fixed large time by its tail supremum. The last term tends to zero by (47). Indeed, its integral

over a fixed finite interval tends to zero, while the remaining integral is bounded by an arbitrarily small L^1 tail. Therefore $q(t) \to \ell_p/A$. Hence

$$\|x(t) - p\|^2 = \frac{E_p(t) - q(t)}{d} \longrightarrow \frac{\ell_p}{A}.$$

Every weak sequential cluster point $\bar{x}$ is a zero of Q. To verify this directly, combine $Q(x(t)) \to 0$ with monotonicity to obtain $\langle Qy, y - \bar{x}\rangle \geq 0$ for every $y \in H$. Set $y = \bar{x} - \varepsilon Q\bar{x}$, divide by $\varepsilon > 0$ and let $\varepsilon \downarrow 0$. Continuity gives $\|Q\bar{x}\|^2 \leq 0$. The distance limits and the Opial argument used in the discrete proof show that there is only one weak cluster point. To pass from sequences to the full trajectory, fix $h \in H$. If $\langle x(t) - \bar{x}, h\rangle$ did not tend to zero, a sequence of times tending to infinity would stay separated from zero. Boundedness supplies a weakly convergent subsequence, whose limit must be $\bar{x}$, a contradiction. This proves weak convergence of the trajectory; finite-dimensional strong convergence follows. ∎

The orders in Theorem 8 cannot be replaced by universal little-$o(1/t)$ estimates. The example in [5, Eq. (2.18)], with $H = \mathbb{R}^2 \cong \mathbb{C}$, $Qx = ix$, $\alpha = 3$ and

$$x(t) = \frac{1 + it}{t^2} e^{-it},$$

satisfies

$$t^2\|Q(x(t))\|^2 = 1 + t^{-2}, \qquad t^2\|\dot{x}(t)\|^2 = 1 + 4t^{-4}.$$

Thus both scaled norms tend to 1. This continuous obstruction is already present in the cited work; Theorem 8 supplies the matching upper order and weak convergence for general monotone Lipschitz mappings.

7 Scaled limits and weighted summability

The preceding energies give more than bounded scaled norms. They also show that the residual and velocity have the same asymptotic magnitude at the endpoint.

Proposition 7.1. Under the assumptions of Theorem 1, with $\alpha > 2$ and $\theta = \alpha - 1$, the discrepancy r_k in (6) converges strongly to zero, and there is a finite number $L \geq 0$ such that

$$k\|x^k - Tx^k\| \longrightarrow L, \qquad k\|x^{k+1} - x^k\| \longrightarrow L.$$

For every $p \in \operatorname{Fix} T$,

$$L^2 = \lim_{k\to\infty} G_k(p) - c \lim_{k\to\infty} E_k(p), \qquad c = 2A + A^2/d.$$

Under the assumptions of Theorem 8, the continuous discrepancy $r(t)$ in (44) also converges strongly to zero. There is a finite number $L \geq 0$, depending on that trajectory, such that

$$t\|Q(x(t))\| \longrightarrow L, \qquad t\|\dot{x}(t)\| \longrightarrow L,$$

and $L^2 = \lim_{t\to\infty} G_p(t) - c \lim_{t\to\infty} E_p(t)$ for every $p \in S_Q$.

Proof. Here $a = 1$, $b = d = \alpha - 2$ and $A = d + 1$, as in Section 3.1. We first record an elementary decay observation. If a nonnegative sequence h_k satisfies

$$|h_{k+1} - h_k| \le C/t_k, \qquad \sum_{k \ge K} h_k^2/t_k < \infty,$$

then $h_k \to 0$. For $C = 0$ this is immediate. Otherwise, suppose that $h_n \ge \varepsilon$ for infinitely many indices. Set $\delta = \varepsilon/(4C)$ and choose each such n sufficiently large that $1/t_n \le \delta$. Let $m > n$ be the first index with $\sum_{j=n}^{m-1} 1/t_j \ge \delta$. For $n \le k < m$, the preceding partial sum is less than δ, so $h_k \ge 3\varepsilon/4$. Thus the weighted square sum on this block is at least $9\varepsilon^2\delta/16$. Infinitely many disjoint such blocks contradict the assumed summability.

The update (22) gives

$$\|r_{k+1} - r_k\| \le \|x^k - x^{k-1}\| + \frac{d}{t_k}\|r_k\| \le \frac{B + dM}{t_k},$$

where $B = \sup\|s_k\|$ and $M = \sup\|r_k\|$ are finite. Apply the observation to $h_k = \|r_k\|$, using (10), to obtain $r_k \to 0$. Since $E_k(p)$ and $G_k(p)$ have limits,

$$\|s_k\|^2 = G_k(p) - cE_k(p) + A\|r_k\|^2$$

also has a finite limit $L^2 \ge 0$. The exact residual identity

$$t_k(x^k - Tx^k) = -s_{k+1} + \left(\frac{A}{t_k} - 1\right) r_k$$

shows that its norm has the same limit. Finally $t_k/k \to 1$ and $t_{k+1}/k \to 1$ give the stated indexing.

For the continuous problem, boundedness of $t\dot{x}$ and r gives $\|\dot{r}(t)\| \le C/t$. In logarithmic time $\tau = \log(t/t_0)$, the curve $R(\tau) = r(t_0 e^\tau)$ is Lipschitz and satisfies

$$\int_0^\infty \|R(\tau)\|^2 \, \mathrm{d}\tau < \infty.$$

A Lipschitz square-integrable curve converges to zero: otherwise disjoint intervals of one fixed positive length would each contribute a fixed positive amount to this integral. Hence $r(t) \to 0$. The identity $t^2\|\dot{x}\|^2 = G_p - cE_p + A\|r\|^2$ and monotonicity of the two energies prove convergence of the scaled velocity norm. The identity $tQ(x) = -t\dot{x} - r$ gives the same limit for the residual norm. ∎

7.1 A fixed firmly nonexpansive example

The existence of a vanishing scaled residual does not imply the weighted square summability that is often used to prove little-o rates. This distinction remains relevant even for firmly nonexpansive maps. The following example uses the standard direct sum of averaged rotations appearing in Davis and Yin [7]; the calculation below concerns the endpoint polynomial generated by (1), rather than ordinary powers of that operator.

Recall that T is firmly nonexpansive when

$$\|Tx - Ty\|^2 \le \langle Tx - Ty, x - y\rangle \qquad (x, y \in H).$$

Every average $(I + R)/2$ with R nonexpansive has this property.

Proposition 7.2. There exist a real Hilbert space H, a firmly nonexpansive linear map $T : H \to H$ with $\operatorname{Fix} T = \{0\}$, and an initial point $x^0 \in H$ such that the iteration (1), with $\alpha = \sigma = 3$ and $\theta = 2$, satisfies

$$x^k \longrightarrow 0, \qquad k\|x^k - Tx^k\| \longrightarrow 0, \qquad \sum_{k=1}^{\infty} k\|x^k - Tx^k\|^2 = \infty.$$

The map and the initial point are fixed throughout the iteration. The choice of x^{-1} is arbitrary.

Proof. We first calculate a scalar orbit. For $k \ge 0$, put

$$p_k(z) = \frac{2}{(k+1)(k+2)} \sum_{l=0}^{k} (l+1) z^l.$$

These polynomials satisfy $p_0 = 1$, $p_1 = (1 + 2z)/3$ and

$$(k+3)p_{k+1}(z) = \big(k + 1 + (k+2)z\big)p_k(z) - kzp_{k-1}(z).$$

This is precisely (1) for the scalar linear map $Tz = \lambda z$. Summing the geometric series and its derivative gives

$$(1 - z)p_k(z) = -\frac{2z^{k+1}}{k+2} + \frac{2(1 - z^{k+1})}{(k+1)(k+2)(1-z)}. \tag{51}$$

Let $0 < \delta \le 1/128$ and $z = (1 + e^{2i \arcsin \delta})/2$. Then $|1 - z| = \delta$ and $|z|^2 = 1 - \delta^2$. Define

$$m = \left\lceil \frac{16}{\delta} \right\rceil, \qquad M = \left\lfloor \frac{1}{4\delta^2} \right\rfloor.$$

For $m \le k \le M$, Bernoulli's inequality gives

$$|z|^{2(k+1)} \ge 1 - (k+1)\delta^2 \ge \frac{1}{2}.$$

The second term of (51) has absolute value at most $1/[4(k+2)]$, since $(k+1)\delta \ge 16$. Therefore

$$|(1 - z)p_k(z)| \ge \frac{\sqrt{2} - 1/4}{k+2} \ge \frac{1}{k+2}.$$

It follows that

$$\begin{aligned} \sum_{k=m}^{M} k|(1-z)p_k(z)|^2 &\ge \frac{1}{4} \sum_{k=m}^{M} \frac{1}{k} \ge \frac{1}{4} \log \frac{M+1}{m} \\ &\ge \frac{1}{4} \log \frac{1}{68\delta}. \end{aligned} \tag{52}$$

Here $m \le 17/\delta$ and $M+1 > 1/(4\delta^2)$; the chosen range of δ ensures $2 \le m \le M$.

Now take $H = \ell^2(\mathbb{N};\mathbb{C})$, regarded as a real Hilbert space, and for $j \ge 1$ set

$$\delta_j = e^{-(j+8)^2}, \qquad \lambda_j = \frac{1 + e^{2i \arcsin \delta_j}}{2}, \qquad (Tx)_j = \lambda_j x_j, \qquad x_j^0 = \frac{1}{j}.$$

The operator T is the average of the identity and a diagonal unitary map, so it is firmly nonexpansive. Each $\lambda_j \neq 1$, hence $\operatorname{Fix} T = \{0\}$. Moreover, $\|x^0\|^2 = \sum_{j\ge1} j^{-2} < \infty$. The first momentum coefficient vanishes, and the polynomial recurrence proves

$$x_j^k = \frac{p_k(\lambda_j)}{j}.$$

All coefficients of p_k are nonnegative and sum to one. Thus $|p_k(\lambda_j)| \le 1$, while $p_k(\lambda_j) \to 0$ for every fixed j, since $|\lambda_j| < 1$. Dominated convergence gives $x^k \to 0$ in norm.

Abel summation also gives $k|(1-z)p_k(z)| \le 4$ for $|z| \le 1$. Indeed, the coefficients of p_k increase and their largest value is $2/(k+2)$, so the sum of the absolute coefficients of $(1-z)p_k(z)$ is $4/(k+2)$. For each fixed $|z| < 1$, (51) gives $k|(1-z)p_k(z)| \to 0$. Dominated convergence, with summable majorant $16/j^2$, therefore proves $k\|x^k - Tx^k\| \to 0$.

Finally Tonelli's theorem and (52) yield

$$\begin{aligned}\sum_{k=1}^{\infty} k\|x^k - Tx^k\|^2 &= \sum_{j=1}^{\infty} \frac{1}{j^2} \sum_{k=1}^{\infty} k|(1-\lambda_j)p_k(\lambda_j)|^2 \\ &\ge \frac{1}{4} \sum_{j=1}^{\infty} \frac{(j+8)^2 - \log 68}{j^2} = \infty.\end{aligned}$$

This proves all the assertions for one fixed map and one fixed initial vector. ∎

The same map can be realized as a Douglas–Rachford map for two closed subspaces: in coordinate j, take the real axis and the line making angle $\arcsin \delta_j$ with it. Their reflected composition is multiplication by $e^{2i \arcsin \delta_j}$. This is the subspace construction in [7]; only its endpoint-iteration behavior is used here.

For completeness, this same fixed example has no power improvement over $1/k$. At $m_j = \lceil 16/\delta_j \rceil$, the scalar bound gives

$$m_j \|x^{m_j} - Tx^{m_j}\| \ge \frac{m_j}{j(m_j+2)} \ge \frac{1}{2j}.$$

Since $m_j \ge 16e^{(j+8)^2}$, for every $\varepsilon > 0$ we have

$$\limsup_{k\to\infty} k^{1+\varepsilon} \|x^k - Tx^k\| = \infty.$$

This is consistent with the little-$o(1/k)$ limit already proved for the example.

At the acceleration boundary, the results above do not settle whether the scaled limit in Proposition 7.1 is zero for every strictly averaged nonlinear map $T = (1-\lambda)I + \lambda R$, where $0 < \lambda < 1$ and R is nonexpansive. Proposition 7.2 rules out weighted residual-square summability as a general route to that conclusion; it does not provide a counterexample to the little-$o(1/k)$ conclusion itself.

8 Finite dimensional proximal trajectories without point selection

We prove Theorem 9 for the unchanged recurrence (1). Fix throughout this section

$$\alpha > 1, \qquad 0 < \theta < 1, \qquad \gamma > 0, \qquad \sigma > 0, \qquad 1 < p \le 1/\theta.$$

The construction has three parts. A convex conjugate provides a compact-domain objective whose proximal graph has an oscillating tangential component near a segment of minimizers. An exact decomposition of the low-band states, together with a continuous choice of a same-point initialization, makes the actual discrete trajectory track that component. A fixed norm infimal convolution then makes the objective finite and globally Lipschitz without changing any proximal query along the trajectory. All constants and the final objective are fixed before that final iteration is run.

8.1 The convex conjugate and its minimizer segment

Choose $v_0 > 0$ with $2v_0 < e^{-2}$ and a smooth cutoff χ on the positive half-line such that

$$0 \le \chi \le 1, \qquad \chi(v) = 1 \quad (0 < v \le v_0), \qquad \chi(v) = 0 \quad (v \ge L_0 := 2v_0).$$

For $v > 0$, set

$$\rho(v) = \int_0^v \chi(t)\,\mathrm{d}t, \qquad b(v) = \rho(v)^p, \qquad a(v) = b(v)\sin\!\left(\log\log\frac{1}{\rho(v)}\right). \tag{53}$$

For $v \le 0$, put $a(v) = b(v) = 0$. These functions are constant for $v \ge L_0$, with values denoted by a_∞ and $b_\infty > 0$. Near zero,

$$b(v) = v^p, \qquad a(v) = v^p \sin\!\left(\log\log\frac{1}{v}\right), \qquad 0 < v \le v_0. \tag{54}$$

In particular, $a, b = O(v^p)$, $a', b' = O(v^{p-1})$ and $|a''| + |b''| = O(v^{p-2})$ there. Since $p > 1$, their first derivatives extend continuously by zero at the origin, and their second derivatives are integrable.

Let $c = 0$ on $(-\infty, 0]$ and define it on $(0, \infty)$ by

$$\begin{aligned} &c(0) = c'(0) = 0, \\ &c''(v) = |a''(v)| + |b''(v)| + m(v), \qquad m(v) = p(p-1)\chi(v)v^{p-2}. \end{aligned} \tag{55}$$

Thus $c \in C^1(\mathbb{R})$, c'' is nonnegative, integrable and supported in $[0, L_0]$, and

$$M_0 := \int_0^{L_0} c''(v)\,\mathrm{d}v > 0.$$

For $v \geq L_0$, one has $c'(v) = M_0$ and $c(v) = M_0 v - D_0$ for a finite constant D_0. Define

$$F(u,v) = \sqrt{(u - a(v))^2 + b(v)^2} + c(v), \qquad f = F^*. \tag{56}$$

The definition on the whole lower half-plane is $F(u,v) = |u|$; no indicator of a half-plane is added.

Proposition 8.1. The function F is finite, continuous, convex and globally Lipschitz. Its conjugate f is proper, lower semicontinuous and convex, and

$$\operatorname{dom} f = R := [-1,1] \times [0, M_0], \qquad \min f = 0, \qquad \operatorname{argmin} f = S := [-1,1] \times \{0\}. \tag{57}$$

Every value of f on R, including its boundary, is finite. For $v \leq 0$,

$$\partial F(u,v) = \partial |u| \times \{0\}. \tag{58}$$

For $v > 0$, F is differentiable, $|F_u(u,v)| < 1$, and $0 \leq F_v(u,v) \leq M_0$.

Proof. For $v > 0$, write $A = u - a(v)$ and $r = (A^2 + b(v)^2)^{1/2} > 0$. The second derivative in a direction (h,k) is

$$\begin{aligned} \frac{\|(h - a'k, b'k)\|^2}{r} - \frac{\langle (A,b), (h - a'k, b'k)\rangle^2}{r^3} \\ + \left(-\frac{Aa''}{r} + \frac{bb''}{r} + c'' \right) k^2. \end{aligned} \tag{59}$$

The first two terms have nonnegative sum, and the last coefficient is at least $c'' - |a''| - |b''| \geq 0$. This proves convexity on the upper half-plane. On the lower half-plane, $F = |u|$. The estimates $a, b, c = O(v^p) = o(v)$ at zero show that these pieces join continuously and that the restriction to any line has no downward jump of its one-sided derivatives. At a crossing with $u \neq 0$ the derivatives agree; at the origin the two slopes are $-|h|$ and $|h|$ for the tangential component h of the line direction. Thus F is convex on the whole plane. The same directional calculation gives (58), including $v = 0$.

Integration of (55) from zero and, separately, backwards from L_0 gives

$$c'(v) \geq |a'(v)| + |b'(v)|, \qquad M_0 - c'(v) \geq |a'(v)| + |b'(v)| \quad (0 < v < L_0). \tag{60}$$

The gradient formulas are

$$F_u = \frac{A}{r}, \qquad F_v = -a' \frac{A}{r} + b' \frac{b}{r} + c'.$$

They and (60) yield the asserted rectangle bound for the gradient. In the upper tail $F_v = M_0$, and the lower half-plane is covered by (58). Therefore every subgradient of F lies in R, which also proves global Lipschitz continuity.

The support function of R is $h_R(u,v) = |u| + M_0 v_+$. Since a, b are bounded and $c(v) - M_0 v_+$ is bounded, there is $K_0 < \infty$ such that

$$h_R - K_0 \le F \le h_R + K_0.$$

Conjugation reverses these inequalities and $h_R^* = \delta_R$, hence

$$\delta_R - K_0 \le f \le \delta_R + K_0.$$

This proves the exact domain statement, rather than only containment in a bounded set. Since $F(0,0) = 0$, one has $f \ge 0$, and Fenchel duality gives

$$\operatorname{argmin} f = \partial F(0,0) = [-1,1] \times \{0\}.$$

All the remaining assertions are standard conjugacy consequences of the finiteness, continuity and convexity of F. ∎

We record formulas that will be used both for the uniform KL estimate and for the trajectory. For $-1 \le s \le 1$, put

$$t_s = \sqrt{1-s^2}, \qquad h_s(v) = c(v) - s a(v) + t_s b(v). \tag{61}$$

Taking the supremum over u in the definition of F^* yields

$$f(s,y) = \sup_{v \in \mathbb{R}} \{vy - h_s(v)\}. \tag{62}$$

This value formula includes $s = \pm 1$, where the supremum over u need not be attained. For $v > 0$,

$$h_s''(v) \ge m(v), \qquad h_s'(v) \ge \int_0^v m(t)\,\mathrm{d}t \ge p \min\{v, v_0\}^{p-1}. \tag{63}$$

If $z = (s,y) = \nabla F(u,v)$ with $v > 0$, then $|s| < 1$ and

$$u = a(v) + b(v)\frac{s}{\sqrt{1-s^2}}, \qquad y = h_s'(v) = -s a'(v) + t_s b'(v) + c'(v). \tag{64}$$

8.2 A KL inequality uniform near the whole segment

We use the convention that the exponent in $f^q \le C \operatorname{dist}(0, \partial f)$ is q. In the usual KL formulation the associated desingularizing function has power $1 - q$; see [17]. We set $\operatorname{dist}(0, \varnothing) = +\infty$.

Proposition 8.2. For

$$\delta_f = \tfrac{1}{2} p v_0^{p-1}, \qquad C_f = (5p-3)^{1/p},$$

one has

$$f(z)^{1/p} \le C_f \operatorname{dist}(0, \partial f(z)) \tag{65}$$

for every $z \in \operatorname{dom} f$ with $0 < \operatorname{dist}(z, S) < \delta_f$. The estimate is uniform at both endpoints of S as well as in its relative interior.

Proof. Let $z = (s, y) \in R$ with $0 < y < \delta_f$. If $\partial f(z)$ is empty, the assertion follows from the convention above. Otherwise take any $g = (u, v) \in \partial f(z)$. Fenchel duality gives $z \in \partial F(g)$. Equation (58) excludes $v \le 0$, since $y > 0$. Equations (63) and (64) then imply $0 < v < v_0$, uniformly in the tangential coordinate and in the choice of g.

For $0 < v \le v_0$, write $\ell = \log(1/v) > 2$ and $\psi = \log \ell$. Direct differentiation gives

$$a''(v) = v^{p-2} \left[p(p-1) \sin \psi - \frac{2p-1}{\ell} \cos \psi - \frac{\sin \psi + \cos \psi}{\ell^2} \right].$$

Thus $|a''(v)| \le p^2 v^{p-2}$. In this range, $b''(v) = m(v) = p(p-1)v^{p-2}$, so (61) gives, uniformly for $s \in [-1, 1]$,

$$2p(p-1)v^{p-2} \le h_s''(v) \le p(5p-3)v^{p-2}. \tag{66}$$

Using (64) in the Fenchel equality, and then $h_s(0) = h_s'(0) = 0$, gives

$$\begin{aligned} f(z) &= s(a - va') + t_s(vb' - b) + vc' - c \\ &= vh_s'(v) - h_s(v) = \int_0^v t h_s''(t) \, \mathrm{d}t. \end{aligned} \tag{67}$$

Consequently,

$$2(p-1)v^p \le f(z) \le (5p-3)v^p. \tag{68}$$

It follows that $f(z)^{1/p} \le C_f v \le C_f \|g\|$. Since this holds for every $g \in \partial f(z)$, taking the infimum proves (65).

For clarity, the finite-value side points $(\pm 1, y)$, $y > 0$, have empty subdifferential. Indeed, a finite subgradient would have $v > 0$, whereas $|F_u(u, v)| < 1$ on that half-plane. They have therefore not been omitted from the neighborhood. On S the value is zero, and on R the distance to S is exactly y. This proves the stated uniformity on the whole segment. A desingularizing function is $r \mapsto C_f r^{1-1/p}/(1 - 1/p)$. ∎

8.3 Exact low band drift and tail averages

Set $T = \mathrm{prox}_{\gamma f}$. Its outputs lie in the fixed compact rectangle R. To distinguish the algorithm parameters from the functions in (53), put

$$a_0 = \alpha - \theta > 0, \qquad c_0 = 1 - \theta > 0, \qquad t_k = k + \sigma - 1 \quad (k \ge 1).$$

In particular, $t_1 = \sigma > 0$ for every allowed clock. We never divide by the possibly nonpositive quantity t_0.

Proposition 8.3. For an actual trajectory of (1), define, for $k \ge 1$,

$$z_k = Tx^{k-1}, \qquad w_k = \frac{t_k}{a_0} x^k - \left(\frac{t_k}{a_0} - 1 \right) z_k.$$

Let $P_1 = 1$ and $P_{k+1} = (1 + c_0/t_k)P_k$. Then there is a vector W and a sequence $B_k \in R$ such that

$$x^k - z_k = \frac{a_0 P_k}{t_k} W + \frac{a_0}{t_k}(B_k - z_k). \tag{69}$$

More precisely, with

$$\mu_j = \frac{c_0}{t_j P_{j+1}} = \frac{1}{P_j} - \frac{1}{P_{j+1}}, \tag{70}$$

one has

$$\begin{aligned} W &= w_1 - \sum_{j=1}^{\infty} \mu_j z_j = \lim_{k\to\infty} \frac{w_k}{P_k}, \\ B_k &= P_k \sum_{j=k}^{\infty} \mu_j z_j, \qquad \sum_{j=k}^{\infty} \mu_j = \frac{1}{P_k}. \end{aligned} \tag{71}$$

These formulas also hold for the two-state system started from any $z_1 \in R$ and finite w_1. The vector W depends continuously on any continuously parameterized compact family of such starting states.

Proof. The algebra in (7), with $\theta - 1 = -c_0$, gives the exact identities

$$\begin{aligned} x^k &= z_k + \frac{a_0}{t_k}(w_k - z_k), \qquad z_{k+1} = Tx^k, \\ w_{k+1} &= \left(1 + \frac{c_0}{t_k}\right) w_k - \frac{c_0}{t_k} z_k. \end{aligned} \tag{72}$$

Conversely these equations define the equivalent tail dynamics from the stated two-state data. The gamma functional equation gives

$$P_k = \frac{\Gamma(\sigma)}{\Gamma(\sigma + c_0)} \frac{\Gamma(t_k + c_0)}{\Gamma(t_k)} \sim C_\sigma k^{c_0}, \qquad C_\sigma = \frac{\Gamma(\sigma)}{\Gamma(\sigma + c_0)} > 0. \tag{73}$$

Division of the last line of (72) by P_{k+1} and telescoping prove (71), since $z_j \in R$ and $P_k \to \infty$. The weights in B_k are nonnegative and sum to one, so $B_k \in R$. Substituting $w_k = P_k W + B_k$ proves (69).

Each finite iteration depends continuously on its initial state because T is continuous. Moreover, with $C_R = \max_{z\in R} \|z\|$, the tail of the series for W has the uniform bound

$$\left\| \sum_{j=N}^{\infty} \mu_j z_j \right\| \le \frac{C_R}{P_N} \longrightarrow 0.$$

The series is therefore uniformly convergent on every such family. This establishes continuity of W, including the limiting operation over the entire future trajectory. ∎

In particular, if $W = (0, d)$ with $d > 0$, then, writing $q_k = x^k - z_k$, (69) yields

$$q_{k,x} = O(k^{-1}), \qquad q_{k,y} \sim a_0 C_\sigma d\, k^{-\theta} > 0, \qquad q_k \to 0. \tag{74}$$

This is a decomposition of the original dynamics; no forcing term has been added to the algorithm.

8.4 Same point initialization for every positive clock

Proposition 8.4. For every fixed $\sigma > 0$, there is a finite initial point x^0 such that $x^{-1} = x^0$ and the associated vector in (71) is $W = (0, d)$ with $d > 0$.

Proof. For a same-point initialization, the two initial proximal outputs agree, so the first step and first transformed state are

$$\begin{aligned} z_1 &= Tx^0, \qquad x^1 = (1-\theta/\sigma)x^0 + (\theta/\sigma)z_1, \\ w_1 &= Cx^0 + (1-C)z_1, \qquad C = \frac{\sigma-\theta}{\alpha-\theta}. \end{aligned} \tag{75}$$

We treat the vanishing coefficient $C = 0$ separately.

Case $\sigma \neq \theta$. Choose finite $H \in \mathbb{R}$ and $A_* > 0$ so that

$$CH > (|1-C|+1)M_0 + 1, \qquad |C|A_* > |1-C| + 1.$$

When $C < 0$, the height H can and must be chosen negative. For $-A_* \leq \xi \leq A_*$ use the same-point initialization $x^0(\xi) = (\xi, H)$. Since every $z_j \in R$ and the weights μ_j sum to one, (71) and (75) imply

$$W_y(\xi) \geq CH - (|1-C|+1)M_0 > 1, \qquad |W_x(\xi) - C\xi| \leq |1-C| + 1.$$

The endpoint values of W_x have opposite signs, with their order reversed if $C < 0$. The continuity proved in Proposition 8.3 and the intermediate value theorem give a root ξ_* and hence the desired initialization. Although these points need not be in $\operatorname{dom} f$, every proximal query is well-defined on $\mathbb{R}^2$.

Case $\sigma = \theta$. Here (75) forces $x^1 = w_1 = z_1$. Start first with the auxiliary family of tail states

$$z_1(s) = w_1(s) = x^1(s) = (s, M_0), \qquad -1 \leq s \leq 1, \tag{76}$$

and run the two-state system (72) for $k \geq 1$. Its second output is $z_2 = Tz_1$. Let $g = (z_1 - z_2)/\gamma$. We have $g \in \partial f(z_2)$ and $g_y \geq 0$. If $z_{2,y} = M_0$, then $g_y = 0$, and (58) together with $z_2 \in \partial F(g)$ would give $z_{2,y} = 0$, a contradiction. Thus $g_y > 0$, and the upper-half-plane gradient formula gives

$$z_{2,y} < M_0, \qquad -1 < z_{2,x} < 1. \tag{77}$$

By (71),

$$W(s) = \sum_{j=1}^{\infty} \mu_j \big(z_1(s) - z_j(s)\big).$$

All its vertical summands are nonnegative, and the second is strictly positive by (77); hence

$$W_y(s) \geq \mu_2(M_0 - z_{2,y}(s)) > 0.$$

At $s = -1$, all horizontal summands are nonpositive and the second is strictly negative; at $s = 1$ their signs are reversed. Therefore

$$W_x(-1) < 0 < W_x(1).$$

Continuity yields an interior root $s_* \in (-1, 1)$ with $W(s_*) = (0, d)$ and $d > 0$.

It remains to realize this selected tail by a finite original initial point. The auxiliary endpoints were used only for the sign argument and were not asserted to have finite proximal preimages. Choose any finite $V_* > L_0$ and put

$$U_* = a_\infty + b_\infty \frac{s_*}{\sqrt{1 - s_*^2}}, \qquad g_* = (U_*, V_*), \qquad z_* = (s_*, M_0).$$

The constant-tail formulas give $\nabla F(g_*) = z_*$. Therefore $g_* \in \partial f(z_*)$, and the finite same-point initialization

$$x^{-1} = x^0 = z_* + \gamma g_* \tag{78}$$

satisfies $Tx^0 = z_*$. Since $\sigma = \theta$, its first step is exactly $x^1 = w_1 = z_*$, equal to the selected state in (76). The actual original trajectory and that auxiliary tail consequently agree at every subsequent index. This completes both cases. ∎

8.5 Normal bounds and tangential tracking

Fix the actual trajectory selected by Proposition 8.4. Since $z_k \in R$ and $q_k \to 0$, the sequence x^k is bounded. Define the actual proximal graph variables by

$$g_{k+1} = \frac{x^k - z_{k+1}}{\gamma} \in \partial f(z_{k+1}), \qquad z_{k+1} \in \partial F(g_{k+1}). \tag{79}$$

Theorem 7, whose proof uses neither point convergence nor the present construction, gives $g_{k+1} \to 0$ and $x^{k+1} - x^k \to 0$.

Write $z_k = (s_k, y_k)$ and $\eta_{k+1} = g_{k+1,y}$. For all sufficiently large k, $\eta_{k+1} > 0$. Indeed, if $\eta_{k+1} \le 0$, (58) would give $y_{k+1} = 0$, whereas the vertical component of (79) would read

$$\gamma \eta_{k+1} = y_k + q_{k,y} > 0$$

by (74). This contradiction proves the sign. Since $g_{k+1} \to 0$, we also have $\eta_{k+1} < v_0$ eventually. Thus the real trajectory enters the explicitly oscillating branch (54).

Proposition 8.5. For the selected trajectory there are positive constants m_1, M_1 such that, on a tail,

$$m_1 \eta_{k+1}^{p-1} \le y_{k+1} \le M_1 \eta_{k+1}^{p-1}. \tag{80}$$

Moreover,

$$y_k = \Theta(k^{-\theta(p-1)}), \qquad \eta_{k+1} = \Theta(k^{-\theta}), \tag{81}$$

and

$$s_k - \tau_k \longrightarrow 0, \qquad \tau_k = -\frac{\sin(\log\log k + \log\theta)}{\sqrt{1+\sin^2(\log\log k + \log\theta)}} \quad (k \geq 2). \tag{82}$$

Consequently,

$$\omega(z_k) = \omega(x^k) = [-1/\sqrt{2}, 1/\sqrt{2}] \times \{0\}. \tag{83}$$

Proof. For large k, equations (64) and (54) give $|s_{k+1}| < 1$ and

$$g_{k+1,x} = \eta_{k+1}^p \left[\sin\left(\log\log\frac{1}{\eta_{k+1}}\right) + \phi(s_{k+1})\right], \qquad \phi(s) = \frac{s}{\sqrt{1-s^2}}. \tag{84}$$

Also $y_{k+1} = h'_{s_{k+1}}(\eta_{k+1})$. Integrating (66) proves (80); for example, one may take

$$m_1 = 2p, \qquad M_1 = \frac{p(5p-3)}{p-1}.$$

The constants are uniform even when the tangential coordinate approaches an endpoint.

Normal comparison. Put $r = p - 1 > 0$ and $\beta = \theta r$. The vertical component of the actual graph relation is

$$y_{k+1} + \gamma\eta_{k+1} = y_k + Q_k, \qquad Q_k = q_{k,y}. \tag{85}$$

By (74), there are $0 < Q_- < Q_+$ such that eventually

$$Q_- k^{-\theta} \leq Q_k \leq Q_+ k^{-\theta}.$$

Equation (80) implies the two implicit comparisons

$$\begin{aligned} y_{k+1} + \gamma(y_{k+1}/M_1)^{1/r} &\leq y_k + Q_k, \\ y_{k+1} + \gamma(y_{k+1}/m_1)^{1/r} &\geq y_k + Q_k. \end{aligned} \tag{86}$$

Each scalar map $u \mapsto u + Cu^{1/r}$ is strictly increasing on $[0,\infty)$. We verify comparison barriers with valid initial bounds, rather than assuming a stationary balance in (85).

Choose a sufficiently large integer N on the above tail. It has $y_N > 0$. Choose $B > 0$ so small that

$$BN^{-\beta} \leq y_N, \qquad \gamma(B/m_1)^{1/r} \leq Q_-.$$

Then $\ell_k = Bk^{-\beta}$ satisfies

$$\ell_{k+1} + \gamma(\ell_{k+1}/m_1)^{1/r} \leq \ell_k + Q_- k^{-\theta} \leq \ell_k + Q_k.$$

The second comparison in (86) and monotonicity give $y_k \geq \ell_k$ by induction.

For the upper barrier, increase N if necessary and set $A = M_0 N^\beta$, $u_k = Ak^{-\beta}$. Then $u_N = M_0 \geq y_N$. For $k \geq N$,

$$\gamma(u_{k+1}/M_1)^{1/r} \geq \gamma(A/M_1)^{1/r} 2^{-\theta} k^{-\theta}, \qquad u_k - u_{k+1} \leq \beta A k^{-\beta-1}.$$

The exponent $\beta + 1 - \theta = 1 + \theta(p-2)$ is positive. Relative to the displayed lower bound on the dissipation term, the barrier decrement is at most a fixed constant times

$$A^{1-1/r} N^{-\beta-1+\theta} = M_0^{1-1/r} N^{-1},$$

and $Q_+ k^{-\theta}$ is at most a fixed constant times $N^{-\theta}$. For large enough N their sum is at most one. Therefore

$$u_{k+1} + \gamma(u_{k+1}/M_1)^{1/r} \geq u_k + Q_k \qquad (k \geq N).$$

The first comparison in (86) gives $y_k \leq u_k$ by induction. If N was increased, choose the lower constant B at this final value of N. The two barriers prove the first estimate in (81); (80) proves the second. This argument covers every $p > 1$ and does not require $p \leq 2$.

Tangential tracking, including $\theta p = 1$. The normal estimate gives

$$\log\log(1/\eta_{k+1}) = \log\log k + \log\theta + O(1/\log k).$$

Define

$$H_k = \sin(\log\log k + \log\theta), \qquad \lambda_k = \gamma\eta_{k+1}^p = \Theta(k^{-\theta p}), \qquad A_k = \frac{a_0}{t_k}.$$

The condition $\theta p \leq 1$ implies $\sum_k \lambda_k = \infty$ and, on a sufficiently late tail,

$$t_k \lambda_k \geq b_- > 0. \tag{87}$$

At equality, the bound $q_{k,x} = O(k^{-1})$ alone does not give $q_{k,x} = o(\lambda_k)$. We therefore retain its exact future-average structure from (69).

For bounded scalar sequences on this tail, define

$$(\mathcal{M}u)_k = P_k \sum_{j=k}^{\infty} \mu_j u_j.$$

These are probability averages, with exact tail mass

$$P_k \sum_{j=m}^{\infty} \mu_j = \frac{P_k}{P_m} \leq C_0 \left(\frac{k}{m}\right)^{c_0} \qquad (m \geq k \geq N), \tag{88}$$

where the inequality follows from (73). In particular, $(B_k)_x = (\mathcal{M}s)_k$. The target τ_k from (82) is bounded and satisfies, for every fixed $R_* > 1$,

$$\sup_{k \leq j \leq \lceil R_* k \rceil} |\tau_j - \tau_k| \longrightarrow 0, \qquad |\tau_{k+1} - \tau_k| = O((k\log k)^{-1}).$$

For the first assertion, its phase varies by at most $\log(\log\lceil R_* k\rceil/\log k) \to 0$, and both scalar maps defining τ are Lipschitz. Split its probability average at $m = \lceil R_* k \rceil$. The

initial part of $(\mathcal{M}\tau)_k - \tau_k$ tends to zero for fixed R_*, while the tail has absolute value at most $2\sup_j |\tau_j|\, C_0 R_*^{-c_0}$. First letting $k \to \infty$ and then $R_* \to \infty$ proves

$$\kappa_k := |(\mathcal{M}\tau)_k - \tau_k| \longrightarrow 0. \tag{89}$$

This uses slow variation, not convergence of τ_k.

The horizontal components of (79) and (84), together with $q_{k,x} = A_k((\mathcal{M}s)_k - s_k)$, give the exact recursion

$$\begin{aligned} s_{k+1} + \lambda_k \phi(s_{k+1}) &= (1 - A_k)s_k + A_k(\mathcal{M}s)_k - \lambda_k H_k + e_k, \\ e_k &= -\lambda_k \left[\sin\left(\log\log \frac{1}{\eta_{k+1}} \right) - H_k \right], \qquad e_k/\lambda_k \longrightarrow 0. \end{aligned} \tag{90}$$

On a sufficiently late tail $0 < A_k < 1$. Also $\phi'(s) = (1 - s^2)^{-3/2} \geq 1$, so $J_\lambda = (I + \lambda\phi)^{-1} : \mathbb{R} \to (-1, 1)$ is Lipschitz with constant $(1+\lambda)^{-1}$. Compare (90) with $\tau_k = J_{\lambda_k}(\tau_k - \lambda_k H_k)$. Writing

$$d_k = |s_k - \tau_k|, \qquad E_k = \sup_{j \geq k} d_j,$$

we obtain

$$d_{k+1} \leq \frac{(1 - A_k)d_k + A_k E_k + A_k \kappa_k + |e_k|}{1 + \lambda_k} + |\tau_{k+1} - \tau_k|. \tag{91}$$

Here we used positivity and unit mass of the future weights to bound $|(\mathcal{M}s)_k - \tau_k| \leq E_k + \kappa_k$.

Set

$$\rho_k = \frac{A_k + \lambda_k}{1 + \lambda_k}, \qquad r_k = \frac{A_k}{A_k + \lambda_k}.$$

Then $0 < \rho_k < 1$ eventually, $\sum_k \rho_k = \infty$, and (87) gives

$$r_k \leq r_* := \frac{a_0}{a_0 + b_-} < 1.$$

Equation (91) is therefore

$$d_{k+1} \leq (1 - \rho_k)d_k + \rho_k(r_k E_k + \varepsilon_k), \tag{92}$$

where

$$\varepsilon_k = \frac{A_k \kappa_k + |e_k| + (1 + \lambda_k)|\tau_{k+1} - \tau_k|}{A_k + \lambda_k} \longrightarrow 0.$$

Indeed, the first two ratios tend to zero by (89) and $e_k/\lambda_k \to 0$, and the last does so because $A_k + \lambda_k$ is bounded below by a positive multiple of k^{-1}.

The sequence d_k is bounded, so E_k decreases to $D = \limsup_k d_k < \infty$. Given $\epsilon > 0$, the target in (92) is eventually at most $r_*(D + \epsilon) + \epsilon$. Iteration of this scalar inequality, using $\prod(1 - \rho_k) \to 0$, gives $D \leq r_*(D + \epsilon) + \epsilon$. Letting $\epsilon \downarrow 0$ yields $D \leq r_* D$, hence $D = 0$. This proves (82) throughout $\theta p \leq 1$. Only afterwards can one infer

$$(\mathcal{M}s)_k - s_k \longrightarrow 0, \qquad q_{k,x} = o(k^{-1}),$$

by (89) and $|(\mathcal{M}(s-\tau))_k| \le E_k \to 0$. In particular, the critical little-o estimate is a consequence of tracking, not an assumption in its proof.

Finally, the phase $\psi_k = \log\log k + \log\theta$ increases to infinity and has successive increments tending to zero. For each fixed phase modulo 2π, choose the first index crossing that phase plus $2\pi j$; its overshoot tends to zero. Thus the cluster set of $\sin\psi_k$ is exactly $[-1,1]$, and that of τ_k is $[-1/\sqrt{2}, 1/\sqrt{2}]$. Now $y_k \to 0$ and $x^k - z_k \to 0$, proving (83) for the actual iterates. ∎

8.6 Finite Lipschitz extension preserving every proximal query

The preceding compact-domain construction already gives nonselection for the original algorithm. The following fixed infimal convolution improves the function class while preserving that entire trajectory.

Proposition 8.6. Let x^k be the trajectory above, including $k = -1, 0$. There is a fixed $L > 0$ such that

$$\Phi(x) = \inf_{y\in\mathbb{R}^2} \{f(y) + L\|x-y\|\} \tag{93}$$

is finite, continuous, convex, globally L-Lipschitz and coercive, with $\min\Phi = 0$ and $\operatorname{argmin}\Phi = S$. For all actual query indices,

$$\operatorname{prox}_{\gamma\Phi}(x^k) = \operatorname{prox}_{\gamma f}(x^k) \qquad (k \ge -1). \tag{94}$$

The original recurrence with $\operatorname{prox}_{\gamma\Phi}$ and the same initial pair therefore produces exactly the same sequence. Furthermore, there are $\delta_\Phi, C_\Phi > 0$ such that

$$\Phi(x)^{1/p} \le C_\Phi \operatorname{dist}(0, \partial\Phi(x)) \quad \text{if } \operatorname{dist}(x,S) < \delta_\Phi. \tag{95}$$

Proof. For this proof, index graph data by the query input:

$$Y_k = Tx^k, \qquad G_k = (x^k - Y_k)/\gamma \in \partial f(Y_k), \qquad k = -1, 0, 1, \ldots.$$

Since x^k is bounded and $Y_k \in R$, the number

$$B_* := \sup_{k\ge -1} \|G_k\| \tag{96}$$

is finite. Choose once and for all $L > B_*$. In particular, both initialization queries are included, even though $x^{-1} = x^0$ makes them equal. This choice is made from the already constructed trajectory for f and does not assume any property of a trajectory for Φ.

The nonnegativity of f and a trial point in S give

$$0 \le \Phi(x) \le L\operatorname{dist}(x,S) < \infty, \qquad \Phi(x) \ge L\operatorname{dist}(x,R). \tag{97}$$

Joint convexity in (x,y) in (93), or the use of approximate minimizers and their convex combinations, proves convexity of Φ. The triangle inequality gives $|\Phi(x) - \Phi(x')| \le$

$L\|x - x'\|$, hence continuity and lower semicontinuity. The second bound in (97) proves coercivity because R is bounded. If $\Phi(x) = 0$, an approximating sequence y_j in (93) has $y_j \to x$ and $f(y_j) \to 0$. Lower semicontinuity of f implies $f(x) = 0$. This proves $\operatorname{argmin} \Phi = S$.

For any $g \in \partial f(y)$ with $\|g\| \le L$, the subgradient inequality gives, for all x, w,

$$\begin{aligned} f(w) + L\|x - w\| &\ge f(y) + \langle g, w - y\rangle + L\|x - w\| \\ &\ge f(y) + \langle g, x - y\rangle. \end{aligned}$$

Taking the infimum over w and then setting $x = y$ yields

$$\Phi(y) = f(y), \qquad g \in \partial\Phi(y). \tag{98}$$

Apply this to every pair (Y_k, G_k). Proximal optimality gives (94). Starting with the two identical initial queries, induction in (1) proves equality of the two entire trajectories. This argument preserves the selected graph data, not the proximal maps at arbitrary inputs.

It remains to transfer the uniform KL inequality, including points outside the old effective domain. Let $K_L = \{g : \|g\| \le L\}$. Conjugating (93) and changing variables $x = y + q$ gives

$$\Phi^*(g) = F(g) + \delta_{K_L}(g). \tag{99}$$

Since Φ is finite and convex on the whole plane, $\partial\Phi(x)$ is nonempty at every x. It is closed and contained in K_L by the Lipschitz bound. Let g_* be its element of minimum norm. If $\|g_*\| < L$, Fenchel duality and the subgradient sum rule give

$$x \in \partial(F + \delta_{K_L})(g_*) = \partial F(g_*),$$

because F is continuous everywhere and the normal cone of K_L vanishes in its interior. Hence

$$g_* \in \partial f(x), \qquad \Phi(x) = f(x). \tag{100}$$

For $\operatorname{dist}(x, S) < \delta_f$, Proposition 8.2 therefore gives

$$\Phi(x)^{1/p} \le C_f \|g_*\| = C_f \operatorname{dist}(0, \partial\Phi(x)).$$

This also holds trivially on S.

If instead $\|g_*\| = L$, the minimum subgradient norm is exactly L. For $\operatorname{dist}(x, S) < L^{p-1}$, (97) yields

$$\Phi(x)^{1/p} \le \big(L \operatorname{dist}(x, S)\big)^{1/p} \le L = \operatorname{dist}(0, \partial\Phi(x)).$$

Thus (95) holds with $\delta_\Phi = \min\{\delta_f, L^{p-1}\}$ and $C_\Phi = \max\{C_f, 1\}$. Using the minimum-norm subgradient is essential in the second case. The argument does not assert $\Phi = f$

throughout a neighborhood of S; indeed Φ is finite outside R, and the old side-boundary points with empty subdifferential are handled by the second case.

For this construction, $1/p$ is also the smallest possible uniform KL exponent. To see this, take $v \downarrow 0$ and set

$$g_v = (a(v), v), \qquad z_v = \nabla F(g_v) = (0, b'(v) + c'(v)).$$

Then $z_v \to (0,0)$, $\|g_v\| \sim v$, and $g_v \in \partial f(z_v)$. For sufficiently small v, $\|g_v\| < L$, so (98) and (68) imply

$$\Phi(z_v) = f(z_v) \geq 2(p-1)v^p > 0, \qquad g_v \in \partial\Phi(z_v), \qquad \Phi(z_v) \to 0.$$

For any $0 \leq q < 1/p$,

$$\frac{\Phi(z_v)^q}{\operatorname{dist}(0, \partial\Phi(z_v))} \geq \frac{[2(p-1)]^q v^{pq}}{\|g_v\|} \longrightarrow +\infty.$$

Thus no smaller exponent is valid even near $(0,0)$.

∎

Proof of Theorem 9. Use Proposition 8.4 to select a same-point initialization for the prescribed clock. Proposition 8.5 gives its bounded, nonconvergent actual trajectory for f and the exact cluster set. Proposition 8.6 gives one fixed finite, globally Lipschitz, coercive convex objective Φ with the same minimizer set and the stated uniform KL inequality, preserving that trajectory from its initialization onwards. Residual and step convergence follow from Theorem 7, or from their preservation along the original trajectory. The cluster-set conclusion and boundedness imply $\operatorname{dist}(x^k, S) \to 0$; consequently (97) gives $\Phi(x^k) \to 0$. A sequence of finite total length in $\mathbb{R}^2$ is Cauchy and convergent. Since the cluster set is a nondegenerate segment, the total length is infinite. This proves all assertions. ∎

8.7 A standard clock refinement and the scope of the example

For the standard clock $\sigma = \alpha$, the original compact-domain initialization can additionally be chosen in the interior of R, and the Lipschitz constant can be chosen without an unknown trajectory bound. We give the details to distinguish this refinement from the all-clock existence statement.

In this case $C = 1$ in (75), so $w_1 = x^0$. For $x^0 = (\xi, M_0)$, $-1 \leq \xi \leq 1$, the same graph argument as in (77), now at the first output, gives

$$(Tx^0)_y < M_0, \qquad -1 < (Tx^0)_x < 1.$$

The probability weights in (71) then give

$$W_y(\xi, M_0) > 0, \qquad W_x(-1, M_0) < 0 < W_x(1, M_0).$$

The first inequality has a uniform positive margin on the compact top edge, because its lower bound $\mu_1(M_0 - (Tx^0)_y)$ is continuous and strictly positive there. By continuity of W, all three strict inequalities persist for heights $h < M_0$ sufficiently close to M_0. For each such h, the intermediate value theorem selects $\xi_h \in (-1, 1)$ with $W_x(\xi_h, h) = 0$. Thus $x^{-1} = x^0 = (\xi_h, h) \in \operatorname{int} R$ gives the same nonselection conclusion.

Let $D = \operatorname{diam} R = \sqrt{4 + M_0^2}$. For any same-point initialization in R at this standard clock, $W = x^0 - B_1$ gives $\|W\| \le D$. If $\varepsilon_k = a_0 P_k / t_k$, then

$$\frac{\varepsilon_{k+1}}{\varepsilon_k} = \frac{t_k + c_0}{t_k + 1} < 1, \qquad \varepsilon_k \le \frac{a_0}{\alpha}, \qquad \frac{a_0}{t_k} \le \frac{a_0}{\alpha}.$$

Equation (69) therefore gives

$$\|x^k - Tx^k\| \le D\left(1 + \frac{2a_0}{\alpha}\right) < 3D \quad (k \ge 1).$$

At $k = -1, 0$ the residual is at most D. Hence $L = 4D/\gamma$ is a valid fixed choice in (93). For fixed p and γ, this same final objective works for the subfamily $\sigma = \alpha$, $\alpha > 1$ and $0 < \theta \le 1/p$, with an appropriate initial point for each parameter pair. The main theorem for arbitrary σ uses (96) instead; no single-objective assertion over all positive clocks or the whole low band is needed.

The example shows why a KL property of the objective alone cannot supply a general finite-length or point-selection theorem for this time-dependent inertial recurrence. Such conclusions require suitable descent and relative-error estimates for the actual algorithm or an appropriate augmented energy. Theorem 7 supplies residual and step decay but does not supply those additional conditions. In particular, the effective KL exponent $1/p = \theta$ at the critical choice $p = 1/\theta$ still does not guarantee point selection. No conclusion for an effective exponent strictly below θ is asserted. The motion in (82) has an ever-slower phase, rather than a fixed period. The construction is an existence result obtained by continuous shooting; it does not assert an open set of failing initial points or a smooth objective. Coercivity here means $\Phi(x) \to +\infty$ as $\|x\| \to \infty$ and is fully consistent with a segment of minimizers.

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